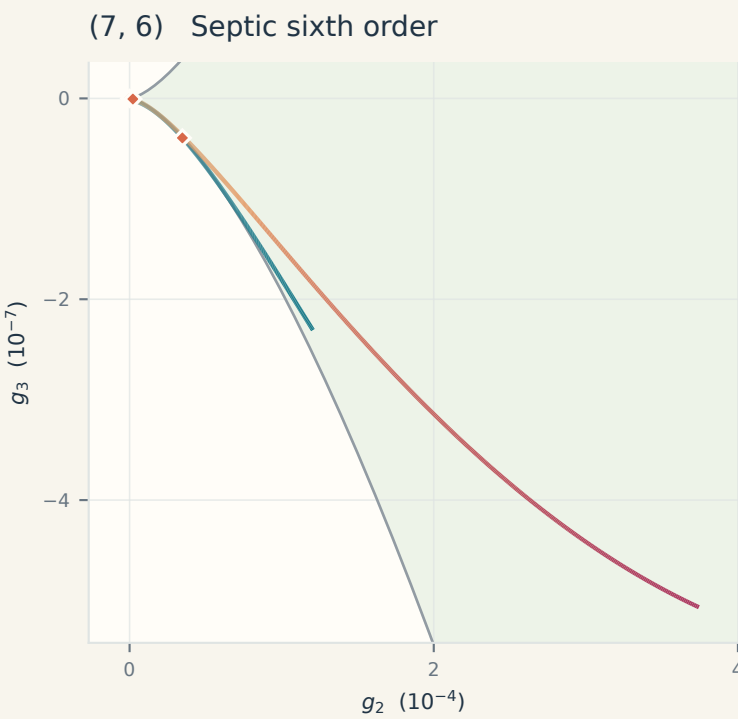
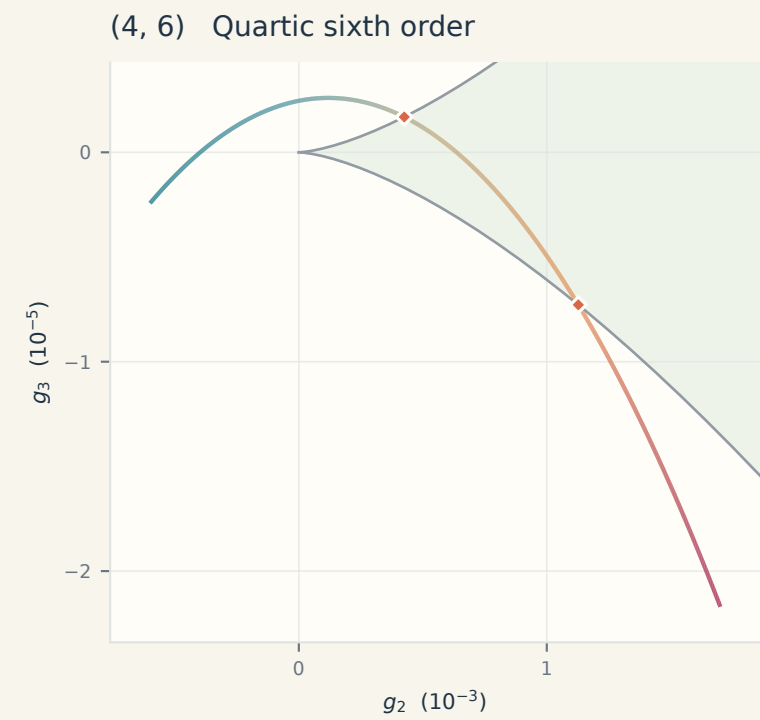
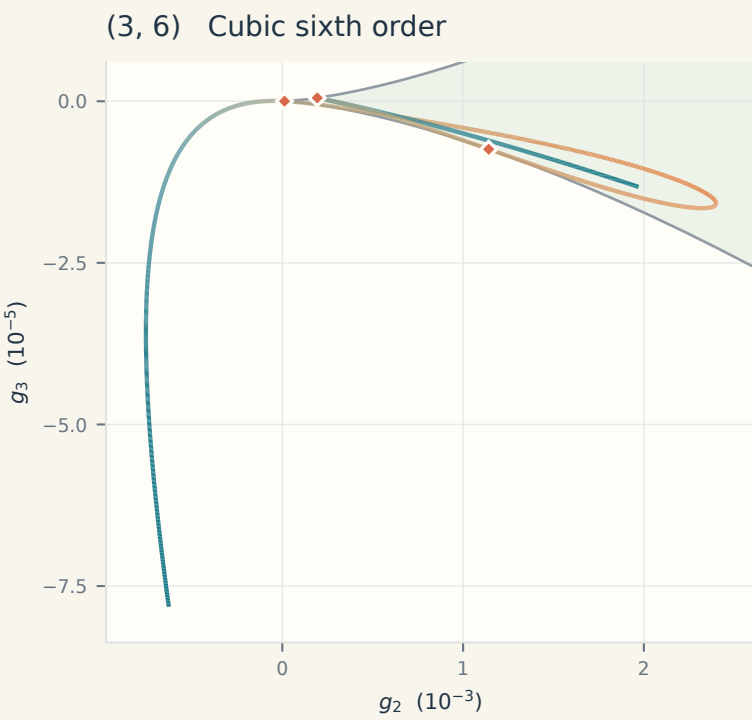
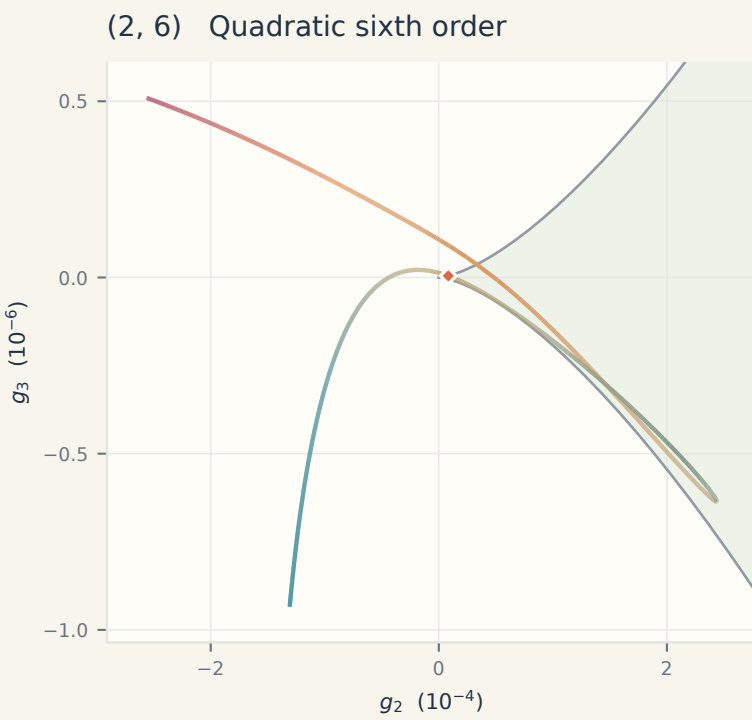
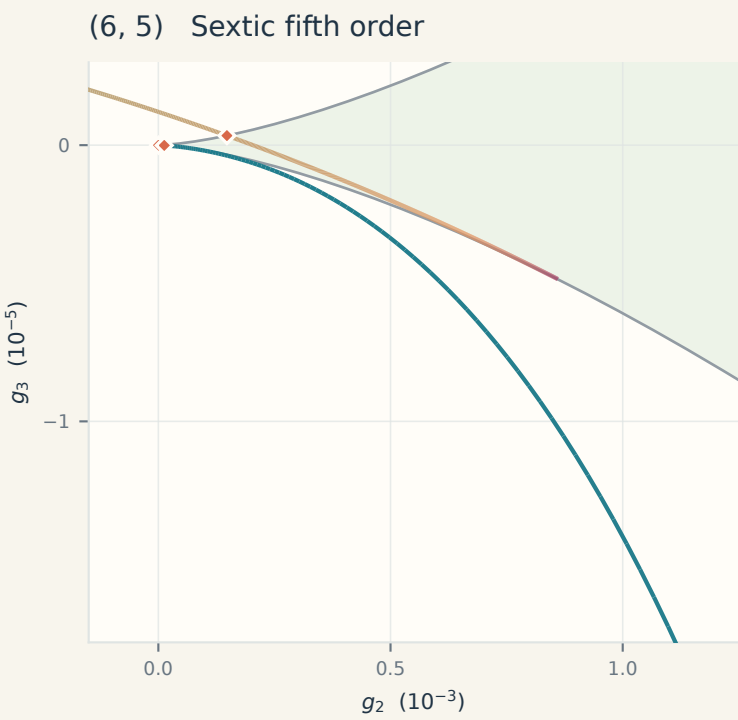
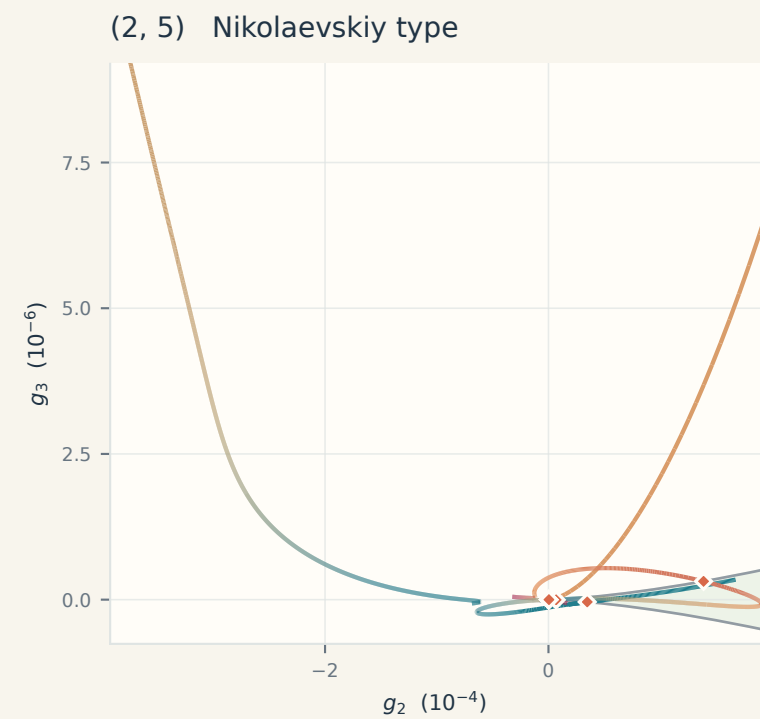
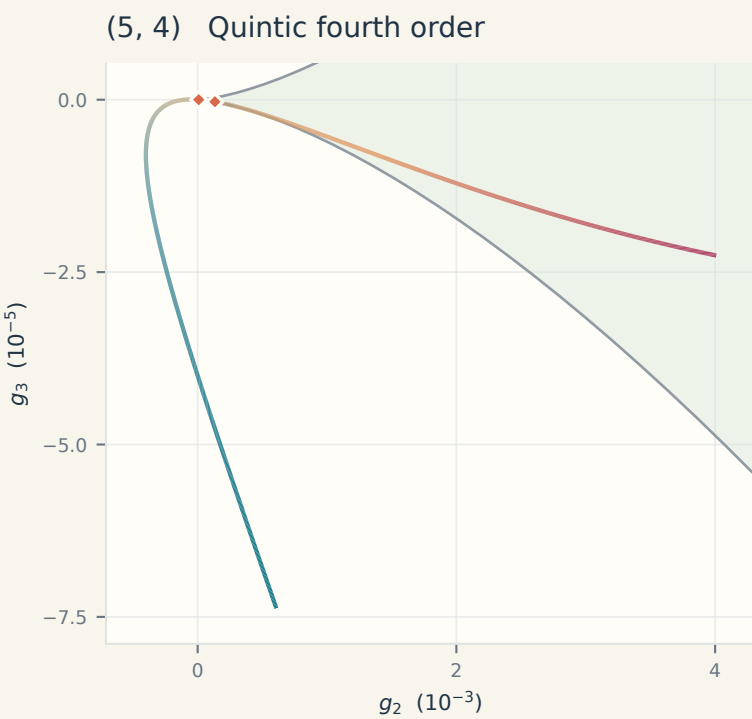
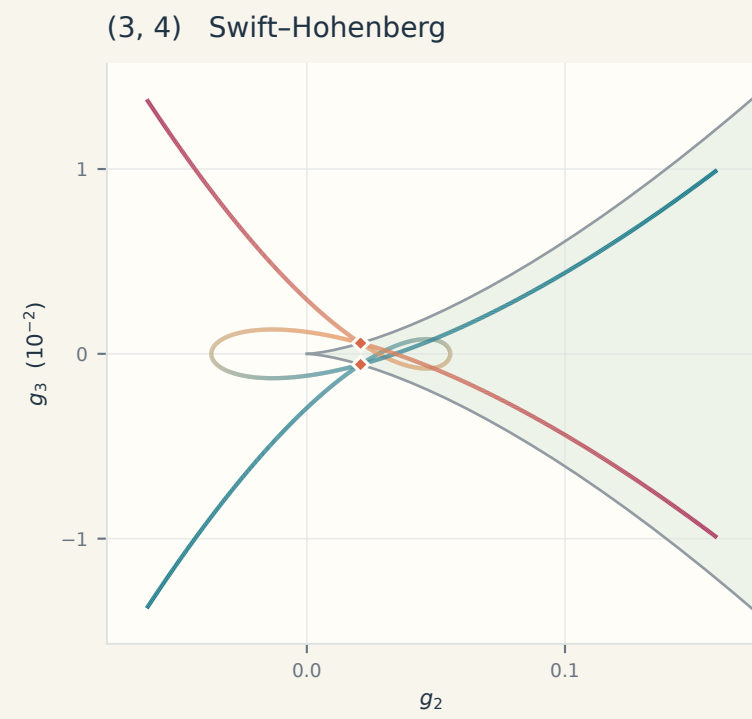
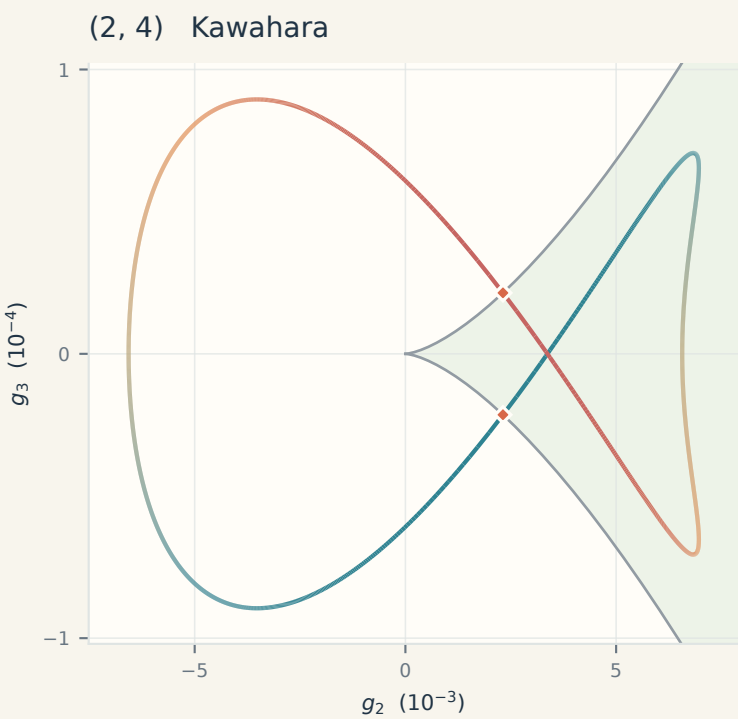
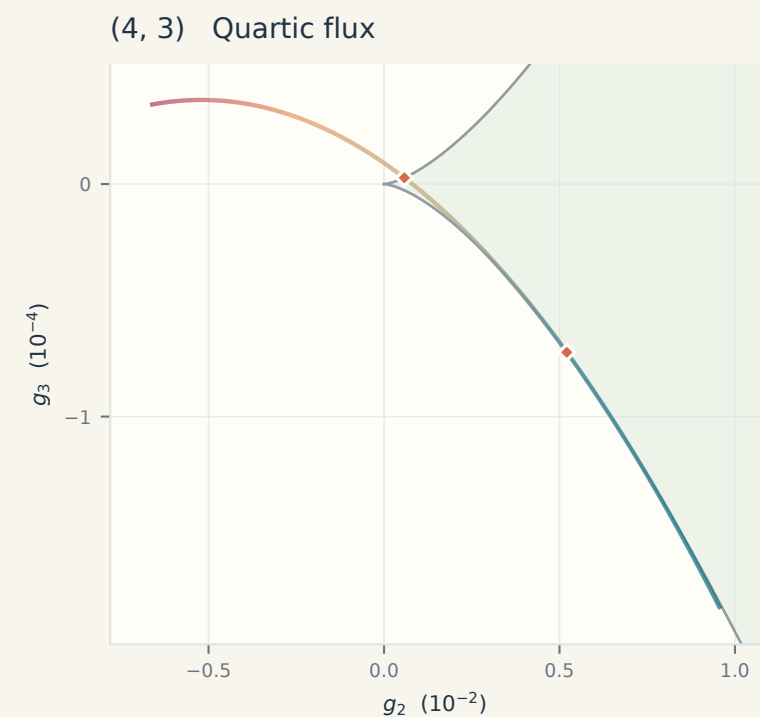
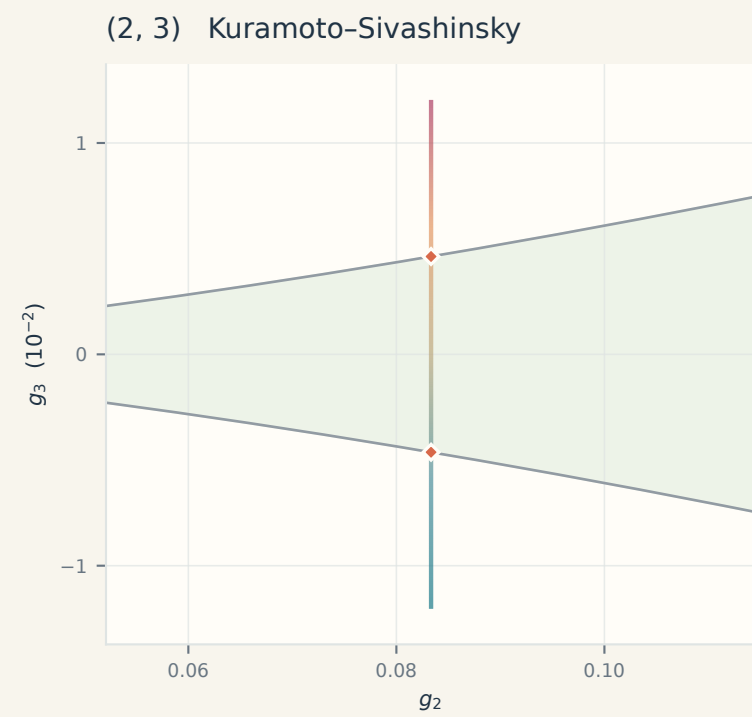
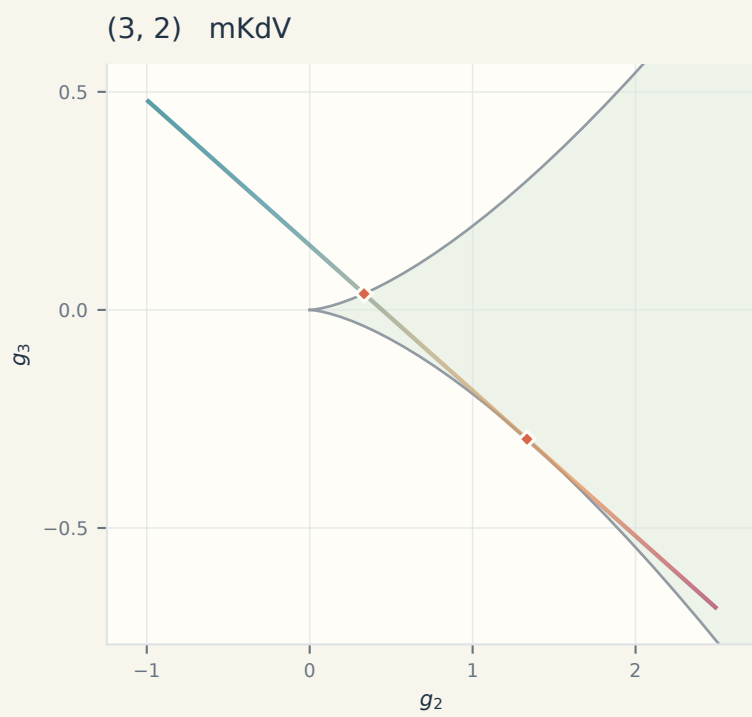
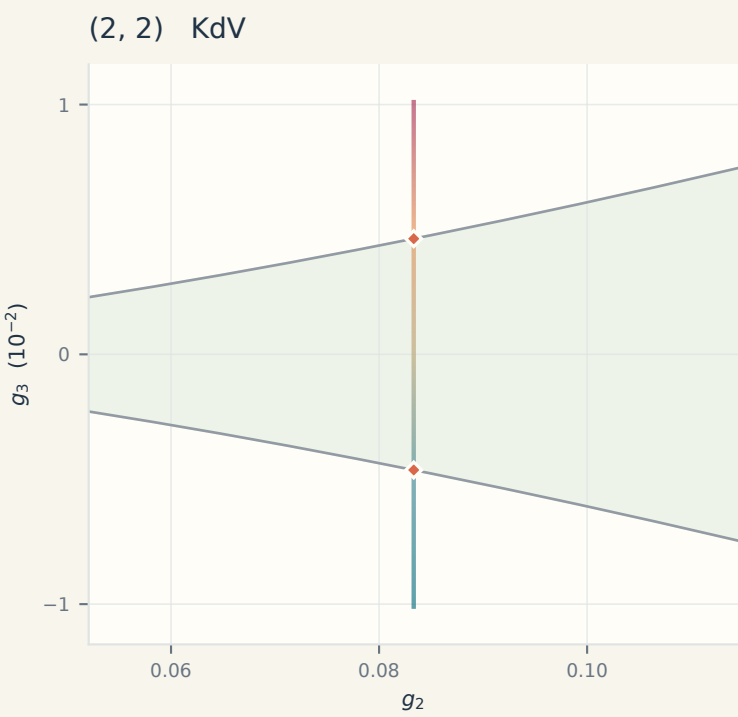


Travelling waves in the plane of lattices

All pole-compatible (n, p) with $2 \leq p \leq 6$ and $2 \leq n \leq 7$



All 30 low-order pairs

$n \setminus p$	2	3	4	5	6
2	2	3	4	5	6
3	1	—	2	—	3
4	—	1	—	—	2
5	—	—	1	—	—
6	—	—	—	1	—
7	—	—	—	—	1

Numbers are pole orders $q = p/(n - 1)$.
Dashes: no integral pole balance.

Reading the plane

$$\Delta = g_2^3 - 27g_3^2$$

Grey cusp: degenerate lattice
Pale interior: $\Delta > 0$
Off the cusp: elliptic lattice
Origin: rational degeneration

Colors follow the parameter
within each declared coefficient slice.

13 pairs · 19 slices

The full book contains every plate,
companion slices, and enlarged details.

[Open index.html](#) for the slider gallery.

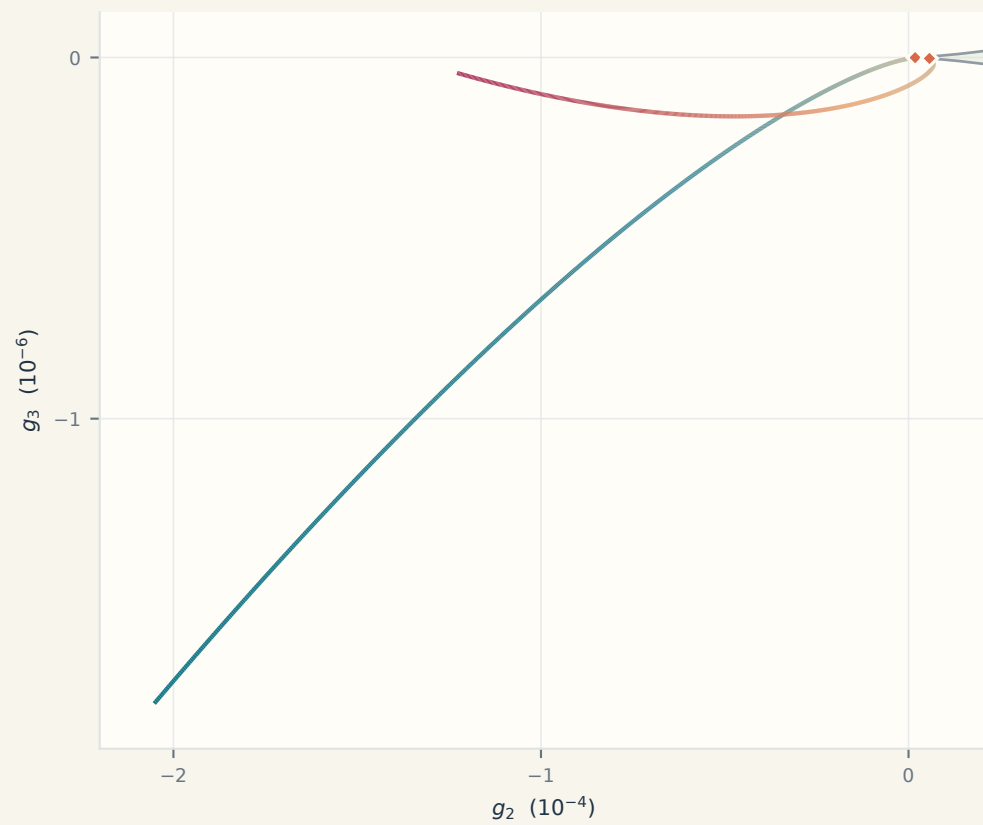
Counts are modulo translation and phase.
A real lattice need not give a smooth,
bounded real-valued wave.

At $p = 2$ the conservative equation
can carry a continuum of waves.

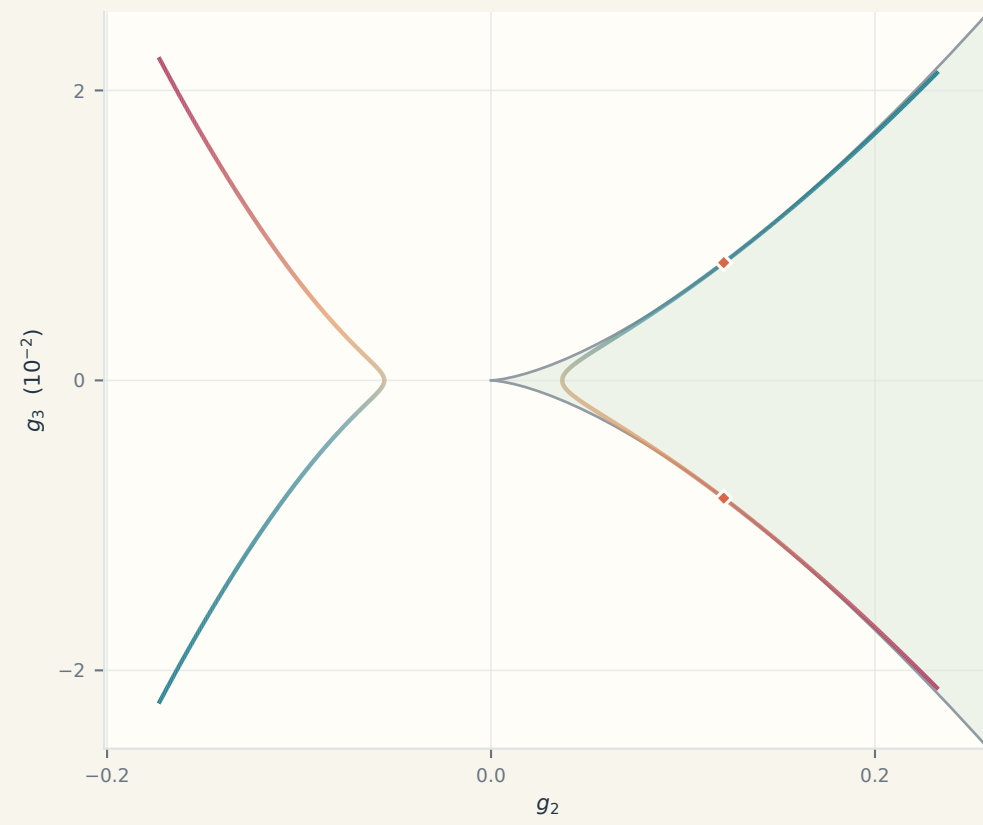
Companion slices · ordinary and twisted branches

Additional normalized coefficient families for the same low-order pairs

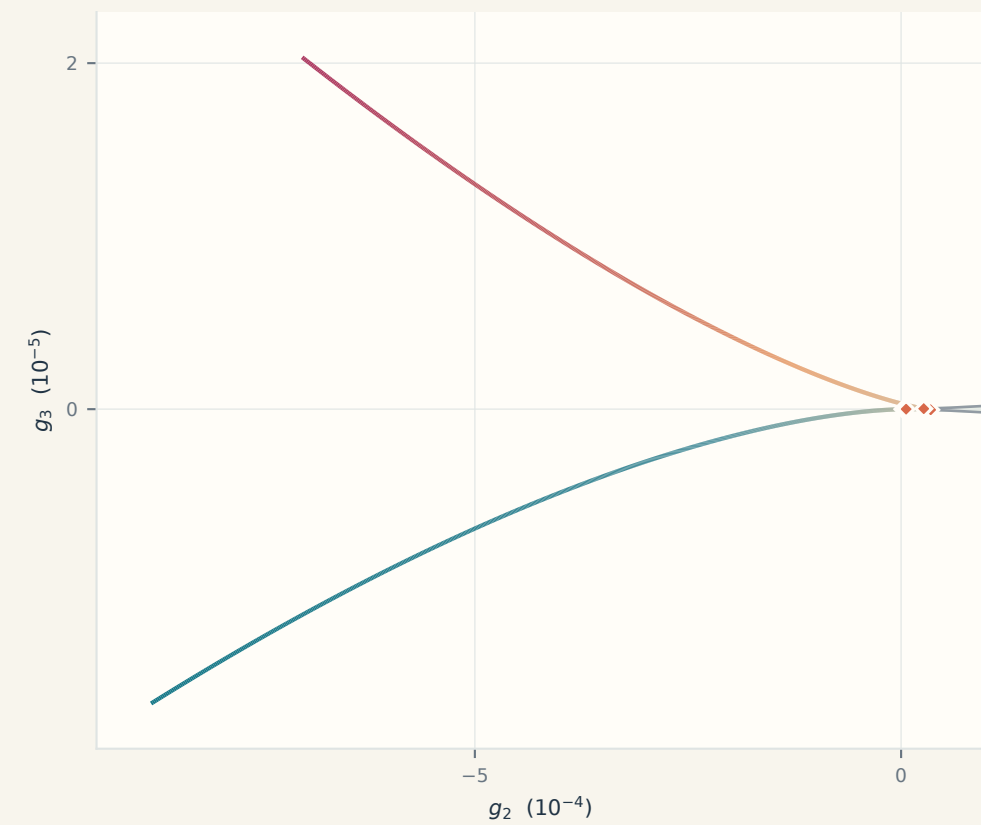
(2, 4) Kawahara-KS



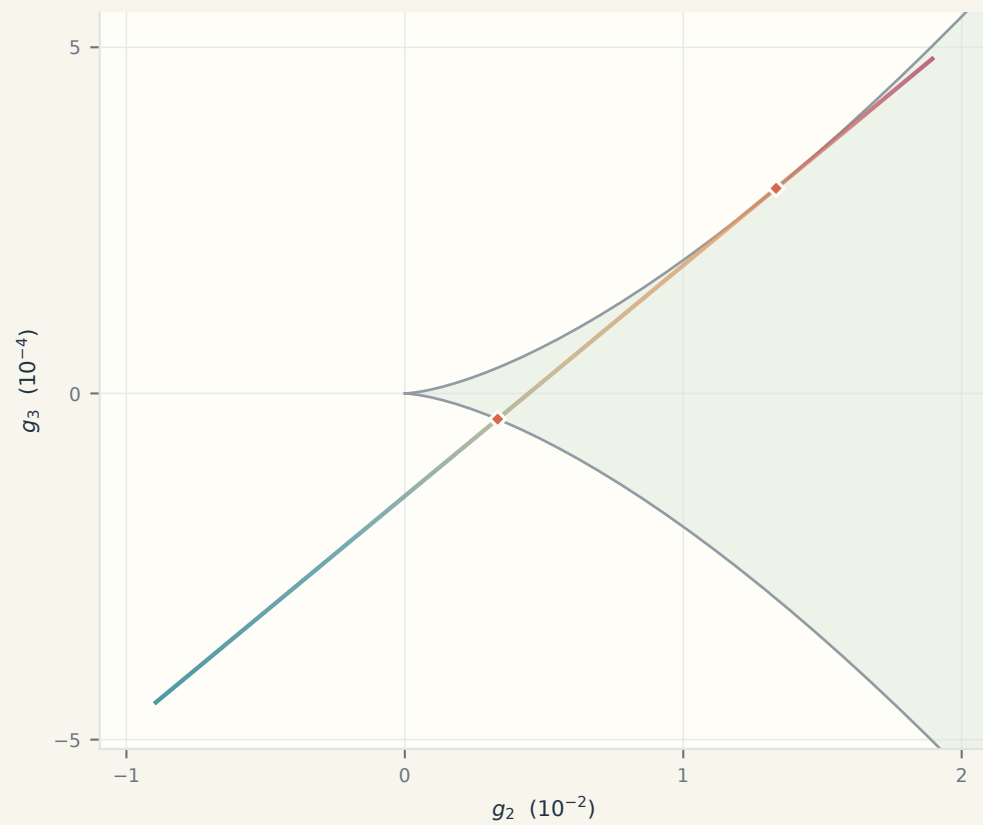
(3, 4) Cubic fourth order



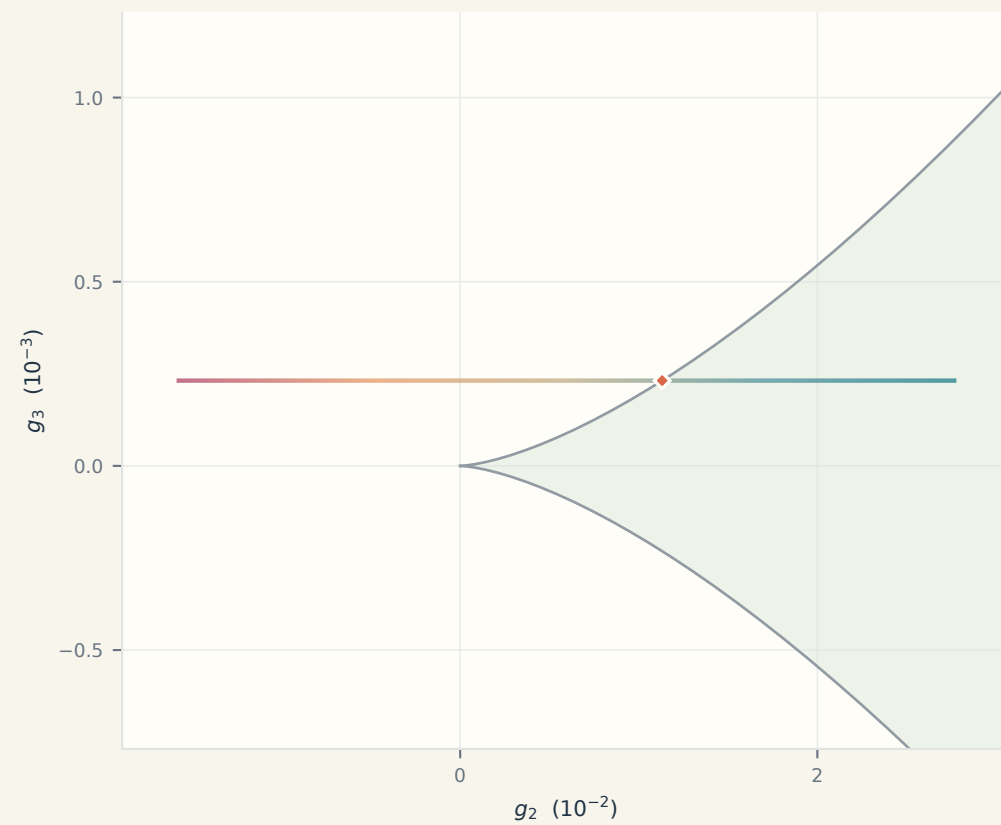
(3, 4) Swift-Hohenberg



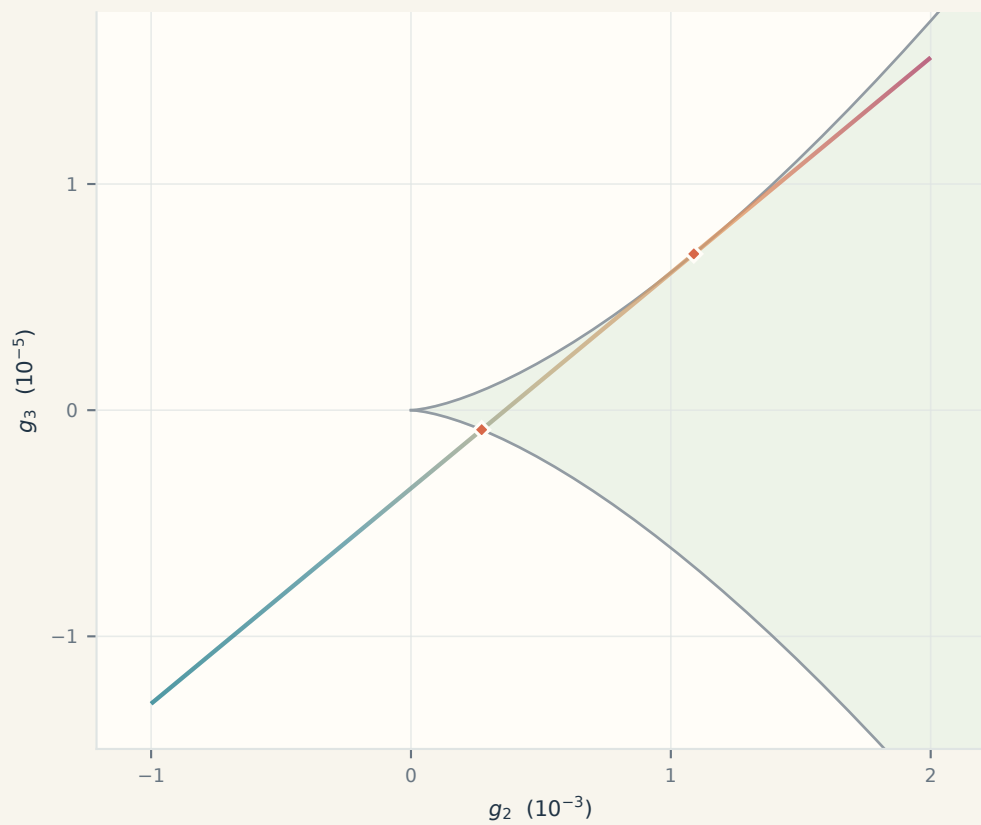
(5, 4) Quintic



(3, 6) Cubic sixth order



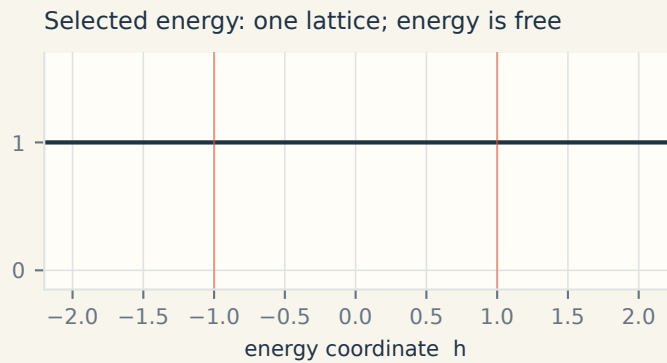
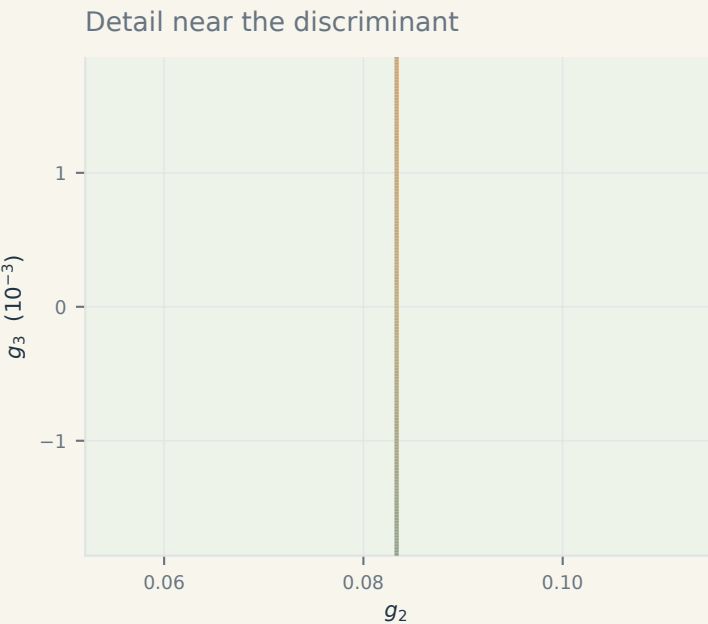
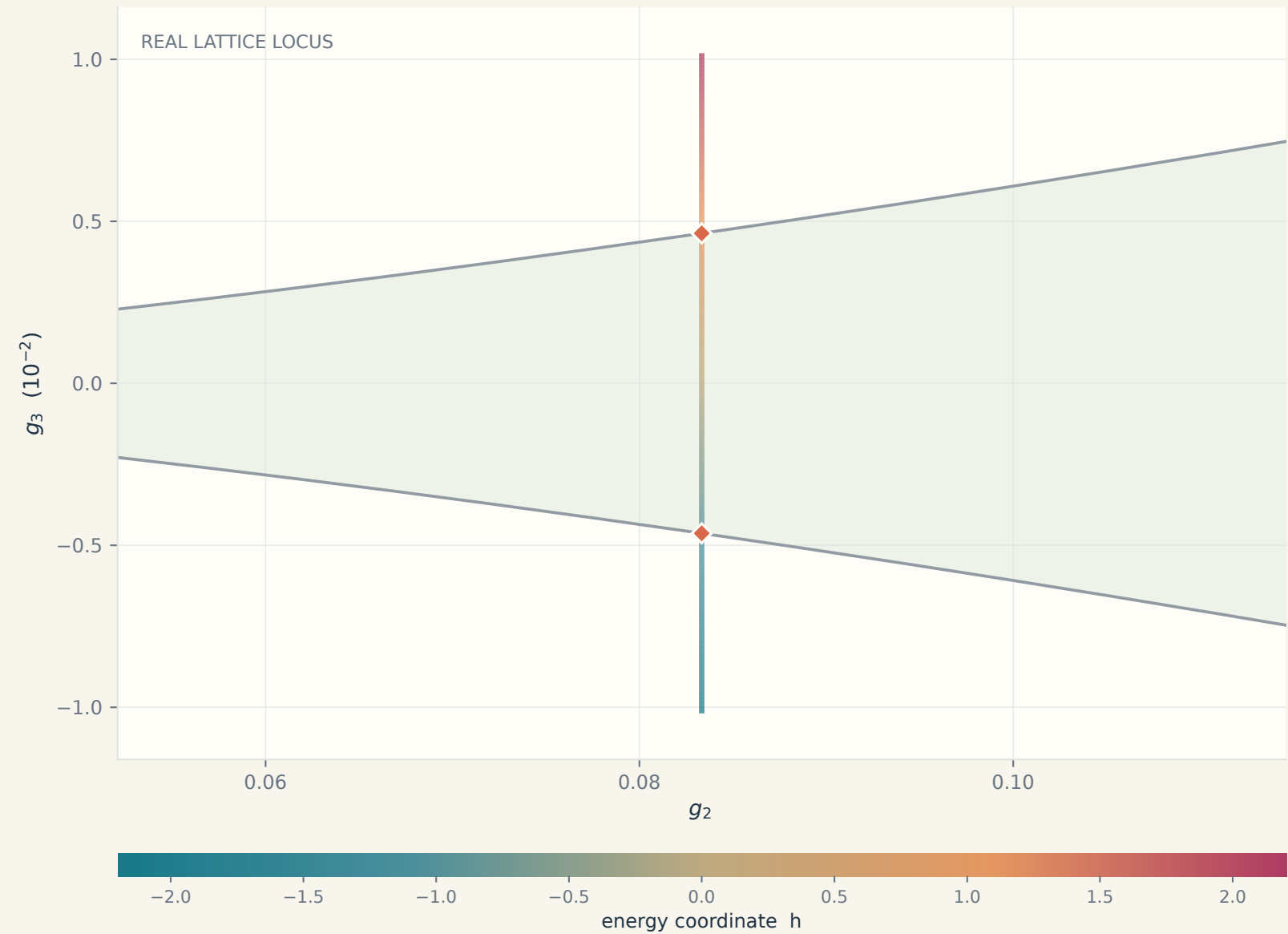
(7, 6) Septic



KdV · the free-energy line

$$u'' - u - 6u^2 = 0$$

$$(n, p) = (2, 2) \cdot q = 2$$



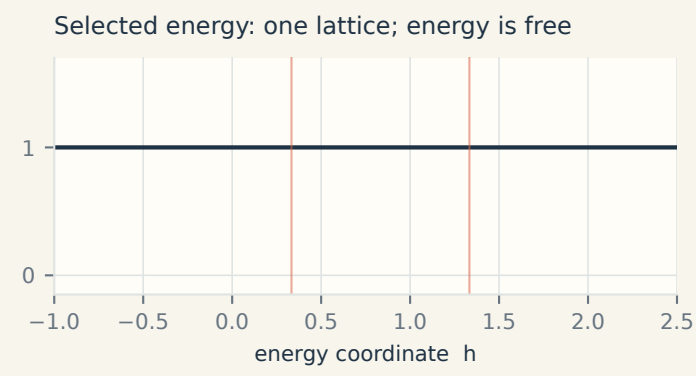
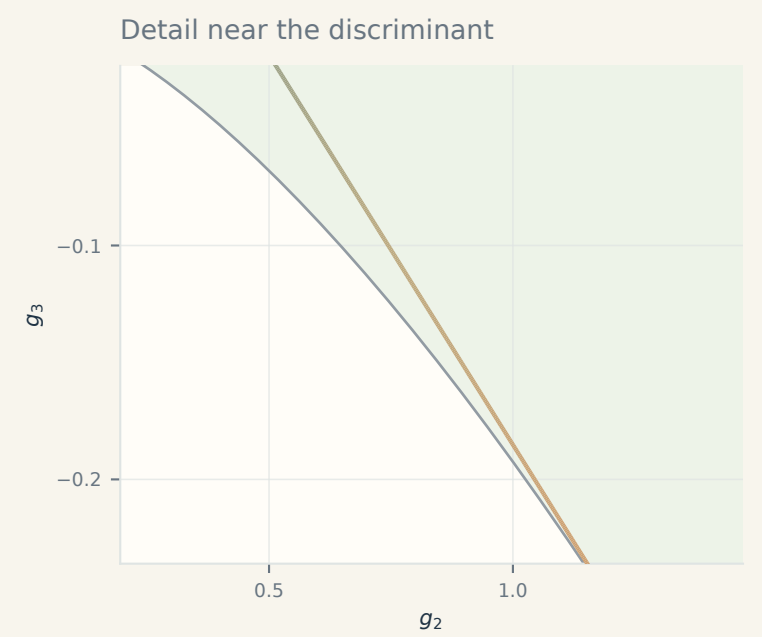
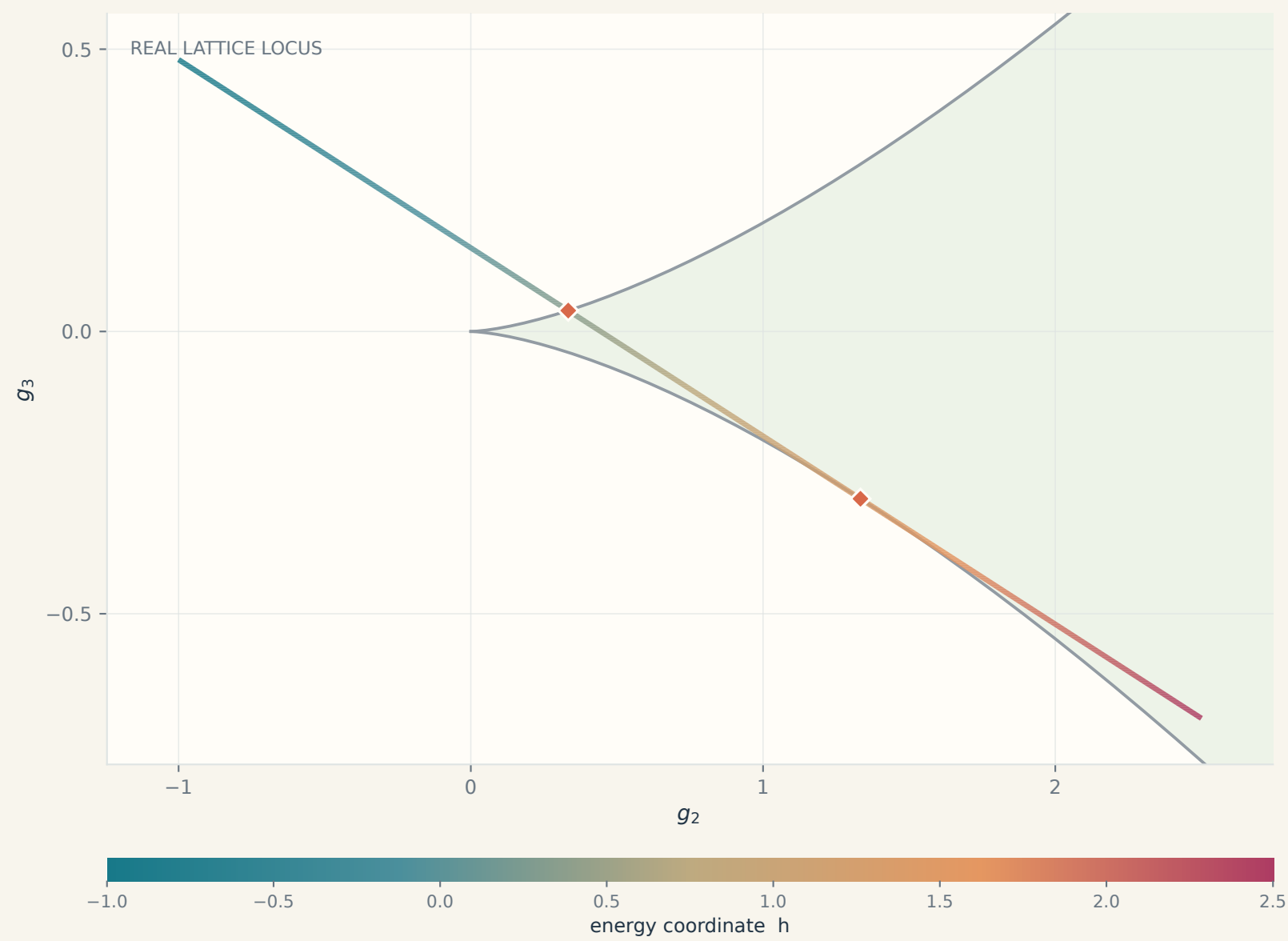
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

One fixed equation carries a continuum of lattices; $h = 216 g_3$ is its first-integral parameter.
The endpoints $h = \pm 1$ are degenerations. The open segment between them has $\Delta > 0$.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

mKdV · a tilted energy line

$u'' - u - 2u^3 = 0$

$(n, p) = (3, 2) \cdot q = 1$



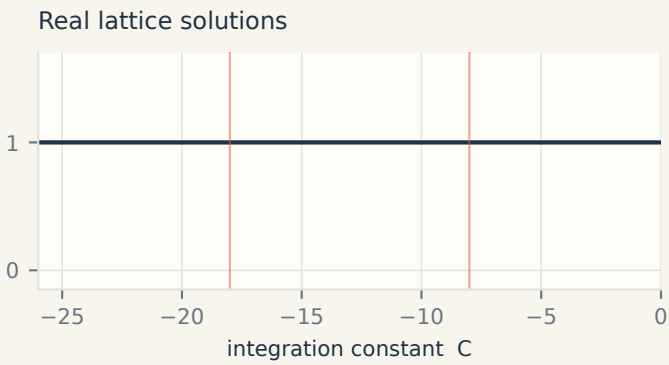
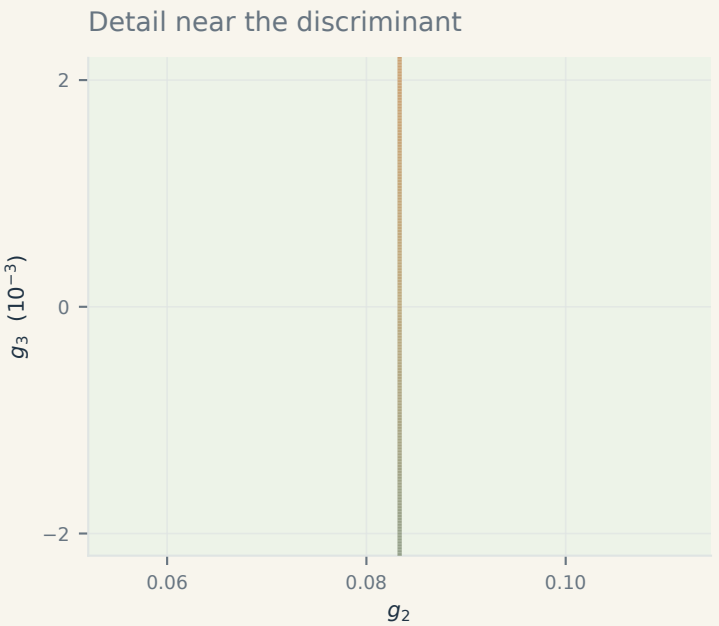
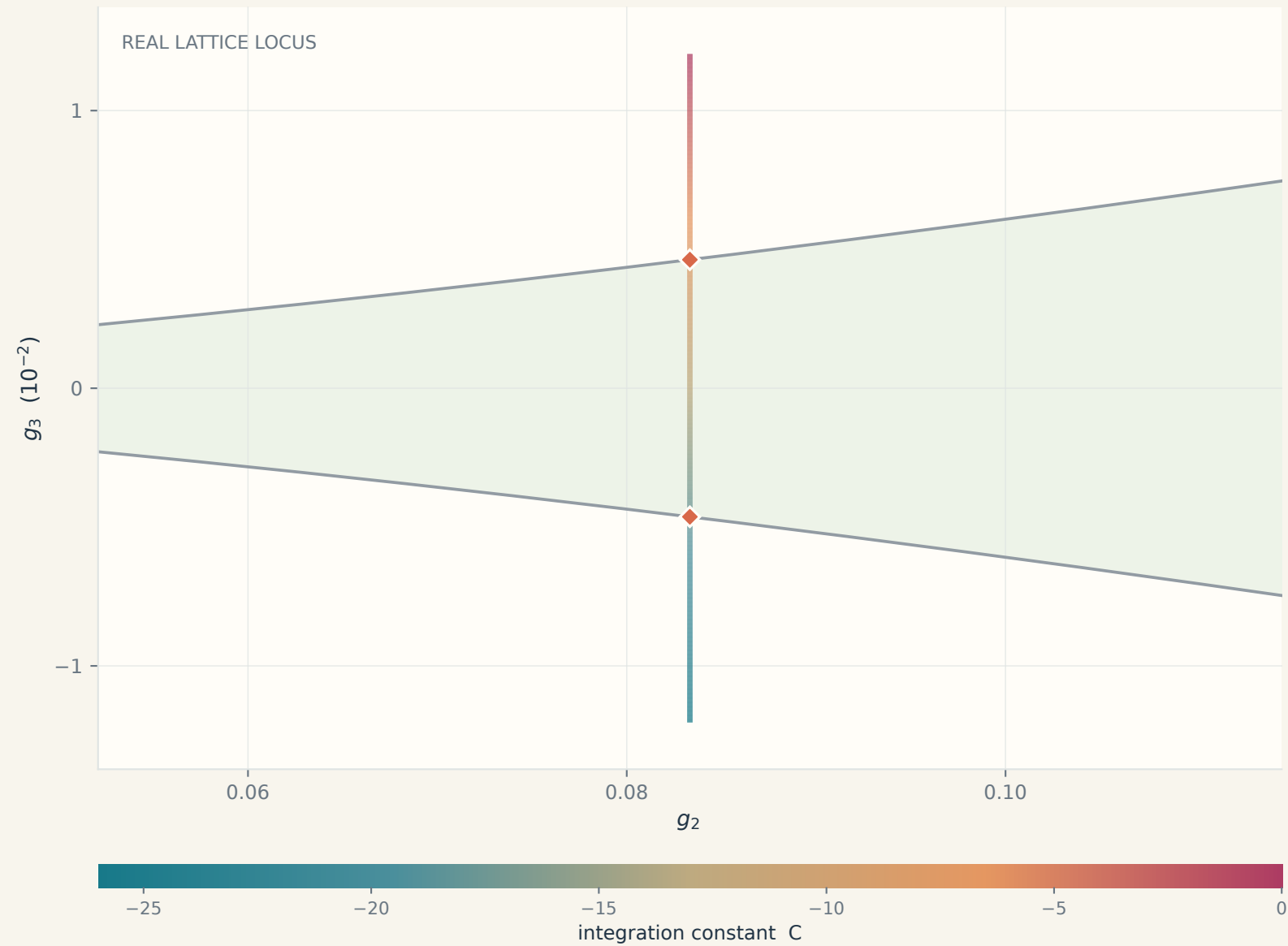
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

Again the equation is fixed: the energy is free. The plotted lattice is the lattice of u^2 .
 $u = \sqrt{(\wp - 1/3)}$ has a character of order two; its ordinary period cell is twice as large.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Kuramoto–Sivashinsky

$$v''' + 4v'' + v' + v^2/2 + C = 0$$

$$(n, p) = (2, 3) \cdot q = 3$$



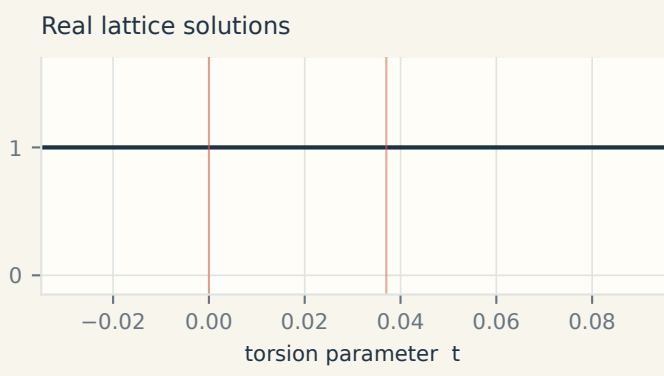
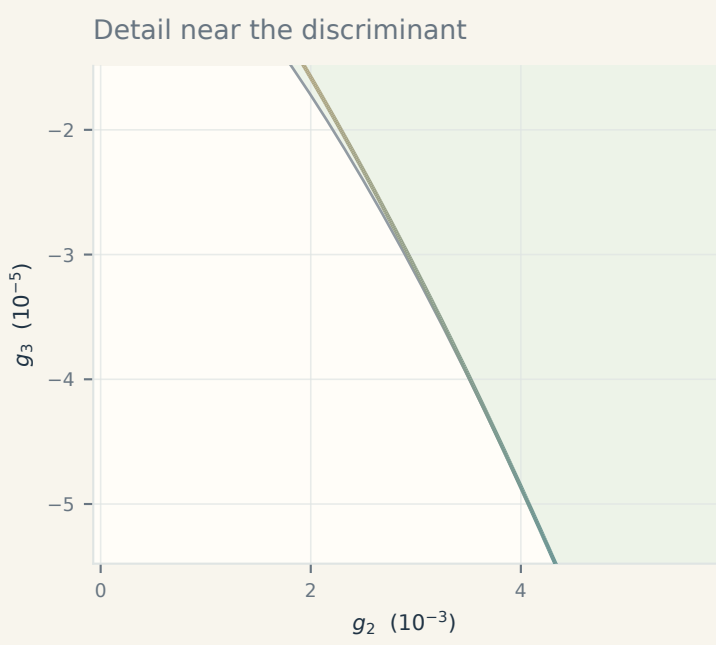
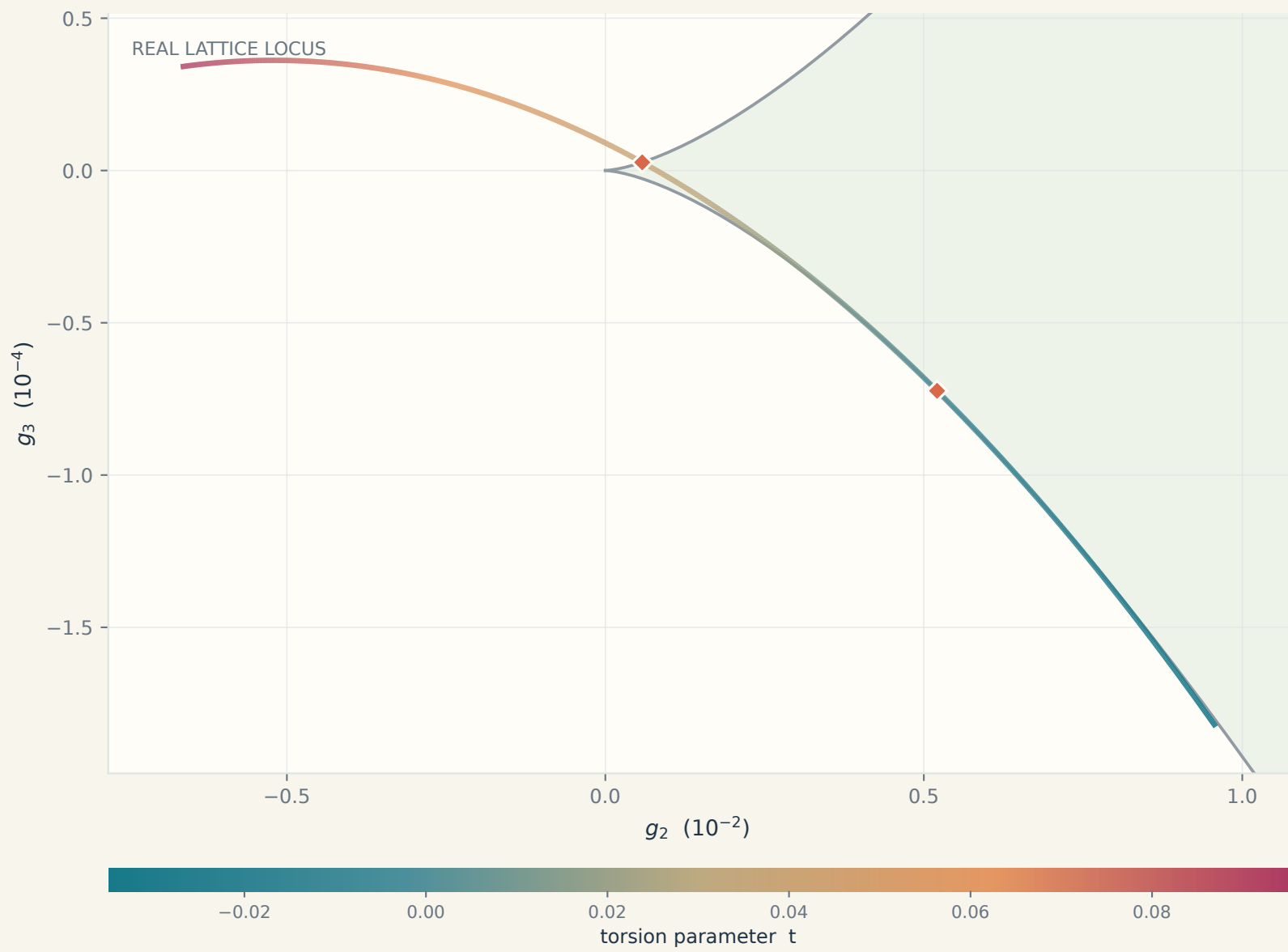
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

One meromorphic wave per fixed C . Elliptic waves require $b^2 = 16a$ in the general KS operator.
 $C = -18$ and -8 are the two degenerations; only the interval between them has $\Delta > 0$.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Quartic flux · a threefold character

$(n, p) = (4, 3) \cdot q = 1$

$P(D)u + 6u^4 = 0, \quad a_2 = 1$



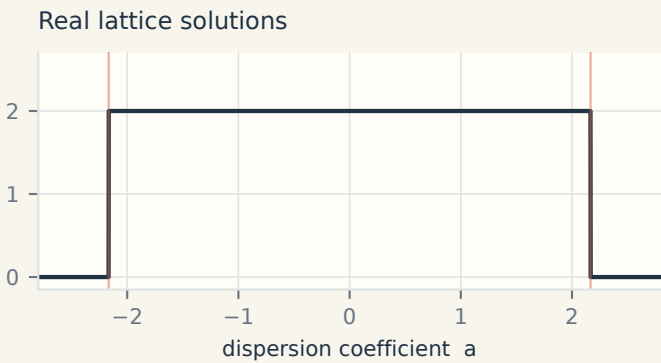
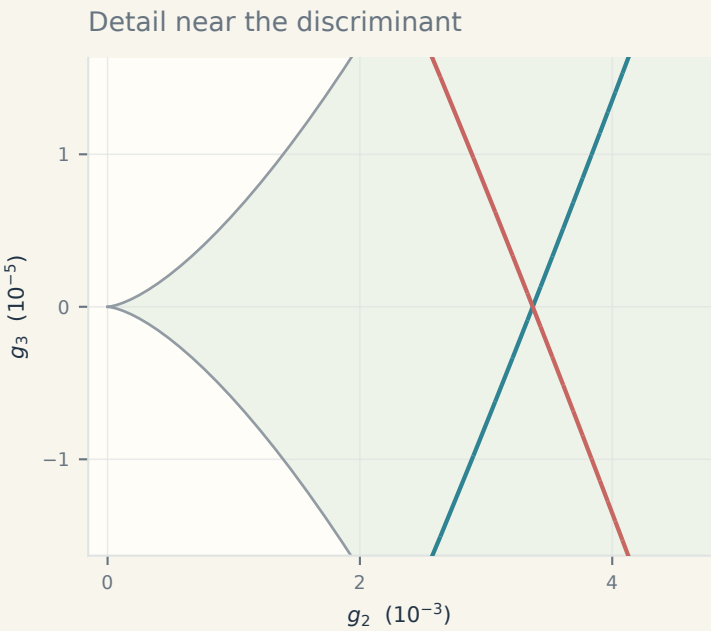
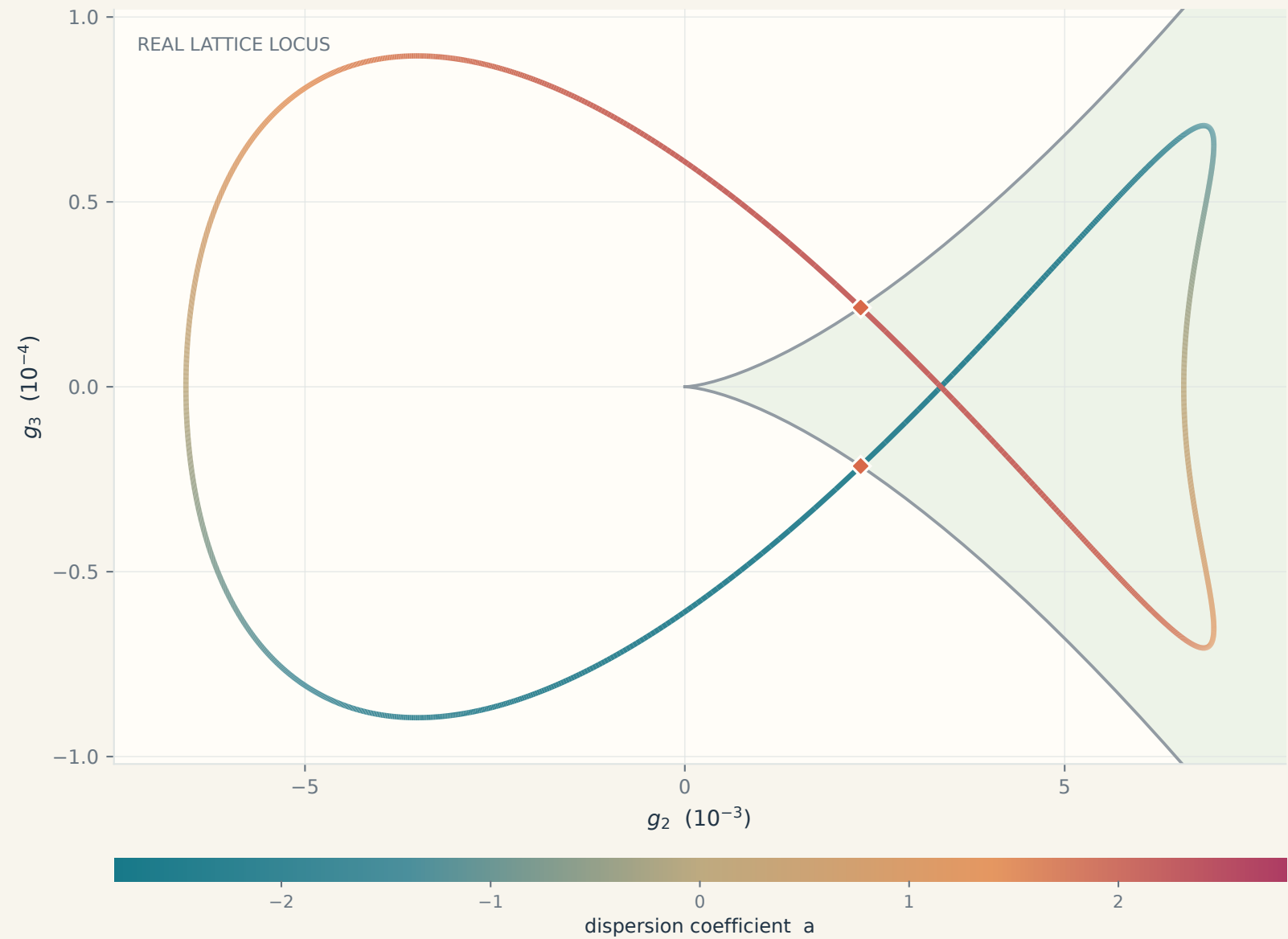
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

The normalized operator is $D^3 + D^2 + D/12 + (54t - 5)/108$; one wave per equation.
A parabola of lattices. Its two real degeneration parameters are $t = 0$ and $t = 1/27$.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Kawahara · the returning loop

$$u'''' + au'' - u - 840u^2 = 0$$

$$(n, p) = (2, 4) \cdot q = 4$$



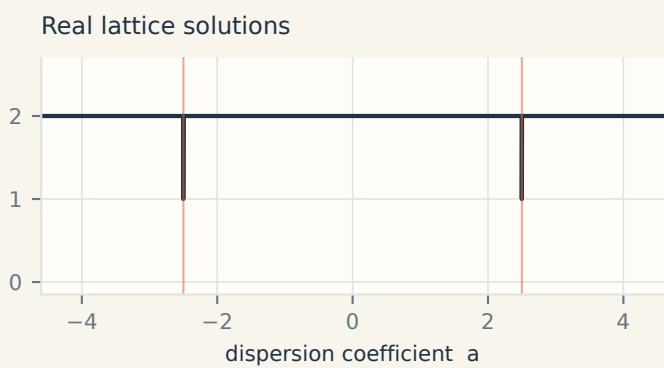
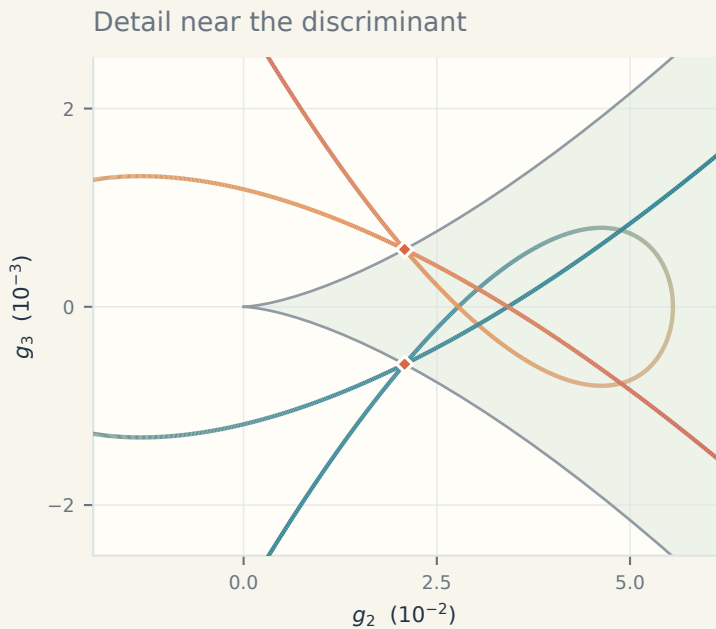
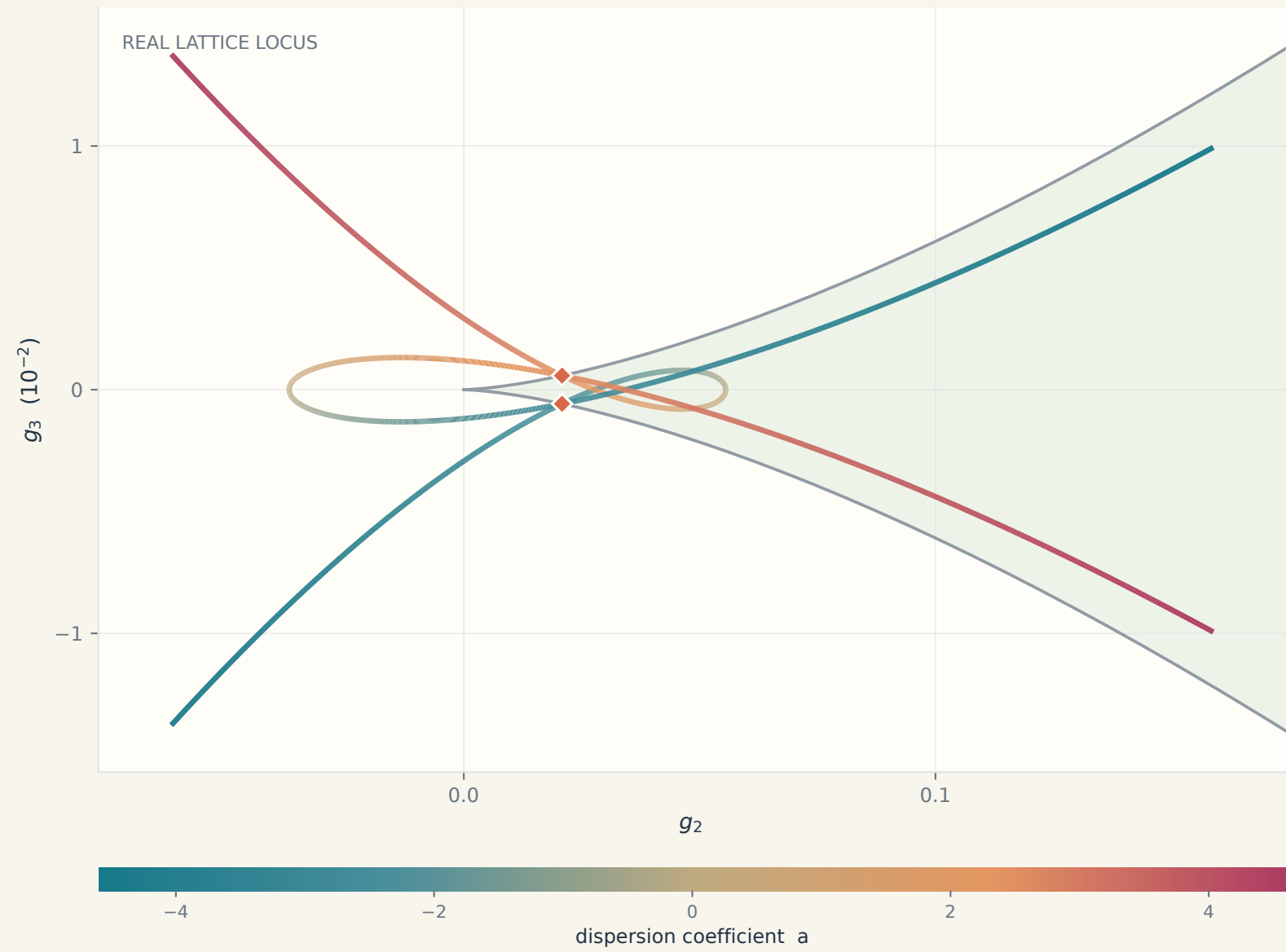
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

Two real lattices for $|a| < 13/6$. At each endpoint they merge into one degenerate wave.
Beyond $|a| = 13/6$ the two lattices are complex conjugates and leave this real plane.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Swift-Hohenberg · two lattice curves

$$u'''' + au'' + u - 120u^3 = 0$$

$(n, p) = (3, 4) \cdot q = 2$



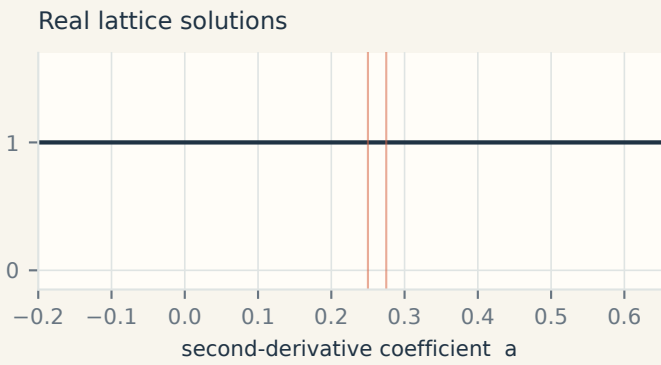
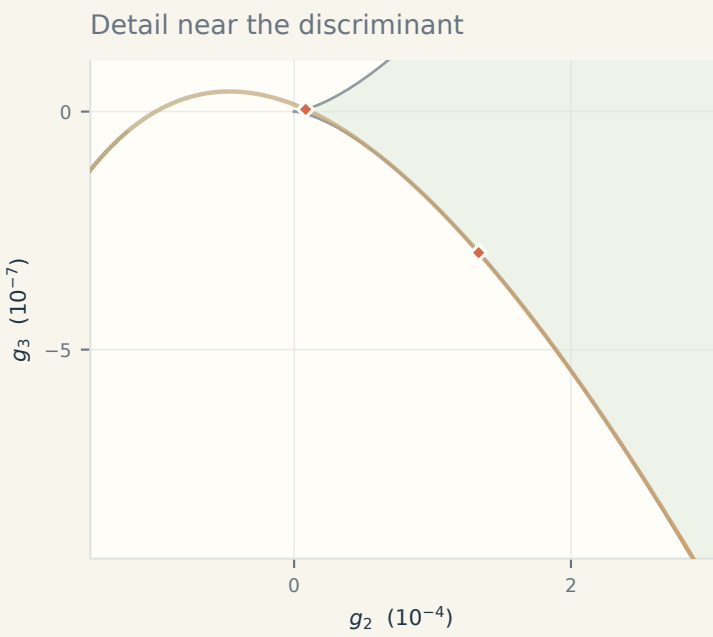
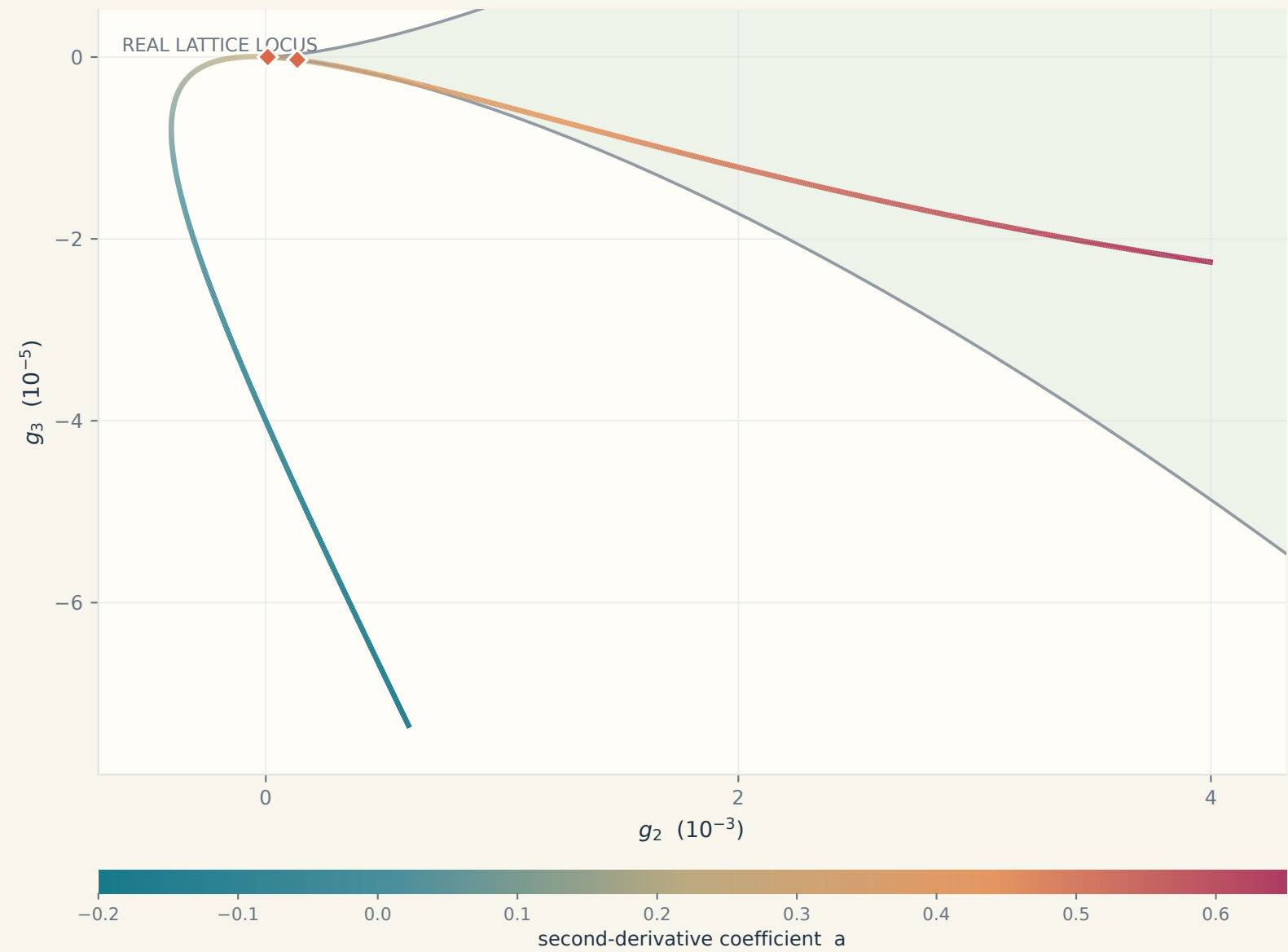
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

Both character branches are shown, up to translation and sign.
They coincide at $a = \pm 5/2$ on the cusp.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Quintic fourth order · a fourfold contact

$(n, p) = (5, 4) \cdot q = 1$

$$u'''' + u''' + au'' + a_1(a)u' + a_0(a)u - 24u^5 = 0$$



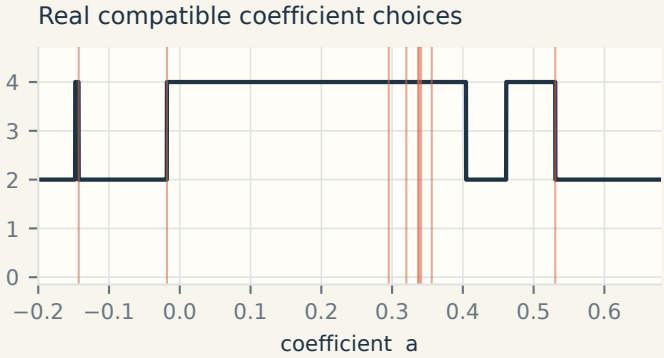
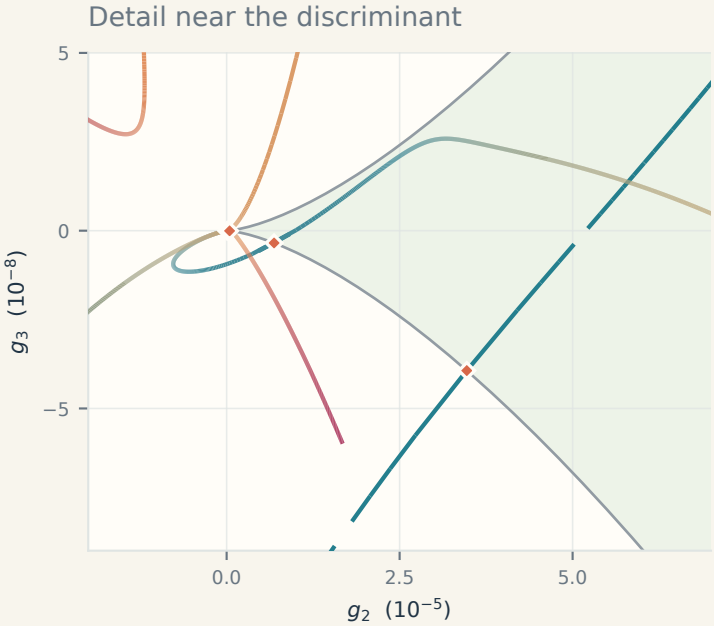
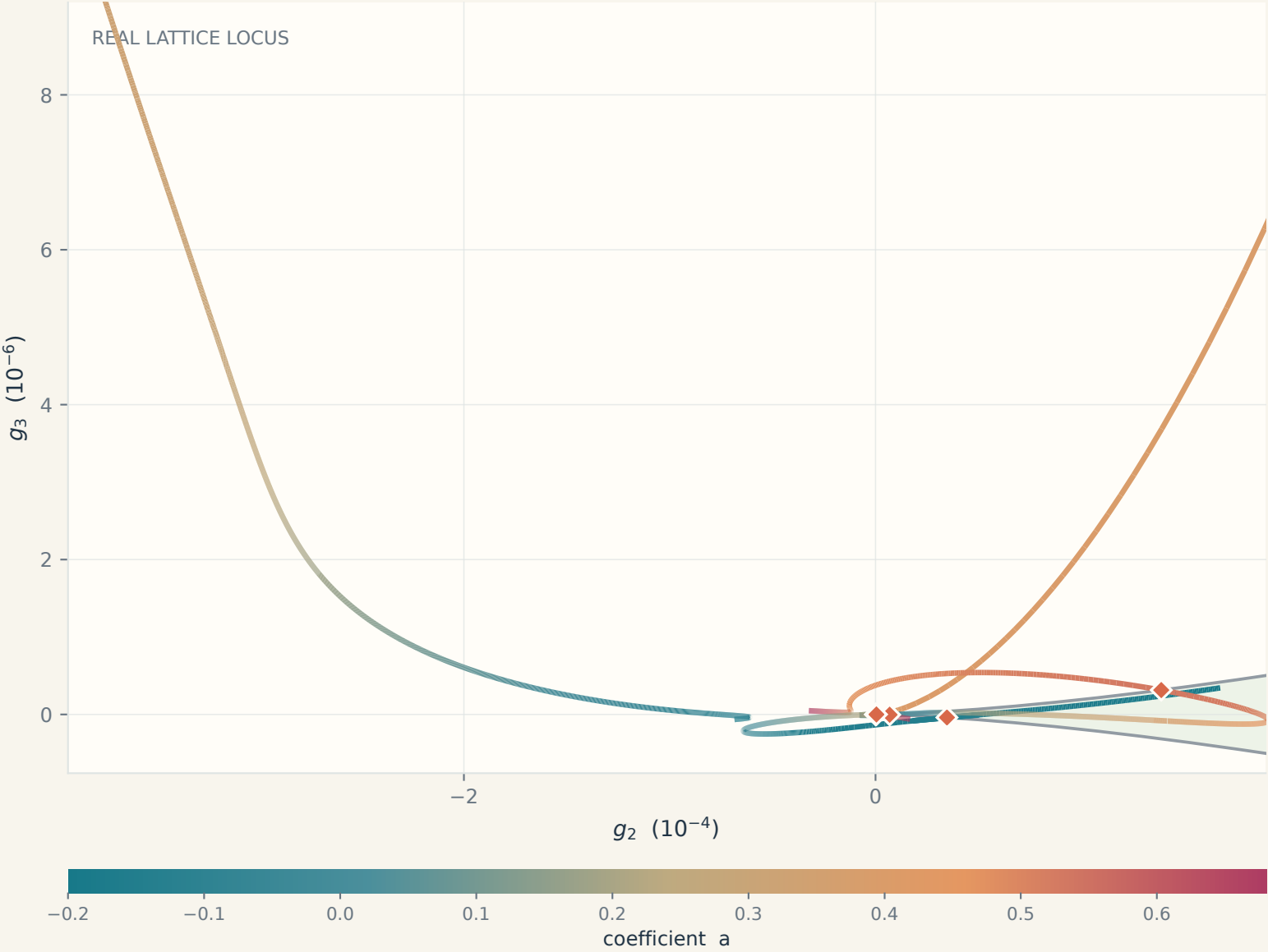
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

One wave per compatible equation, with four ordinary poles and a character of order four.
The contact at $a = 11/40$ has discriminant multiplicity four. The other cusp parameter is $a = 1/4$.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Nikolaevskiy type · four algebraic branches

$$(n, p) = (2, 5) \cdot q = 5$$

$$u^{(5)} + u^{(4)} + au''' + a_2u'' + a_1u' + 15120u^2 + C = 0$$



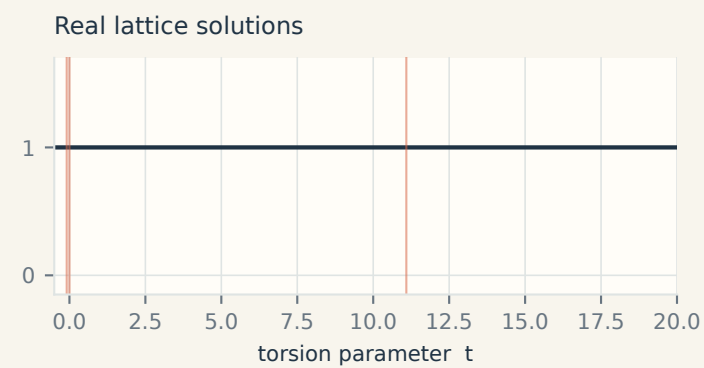
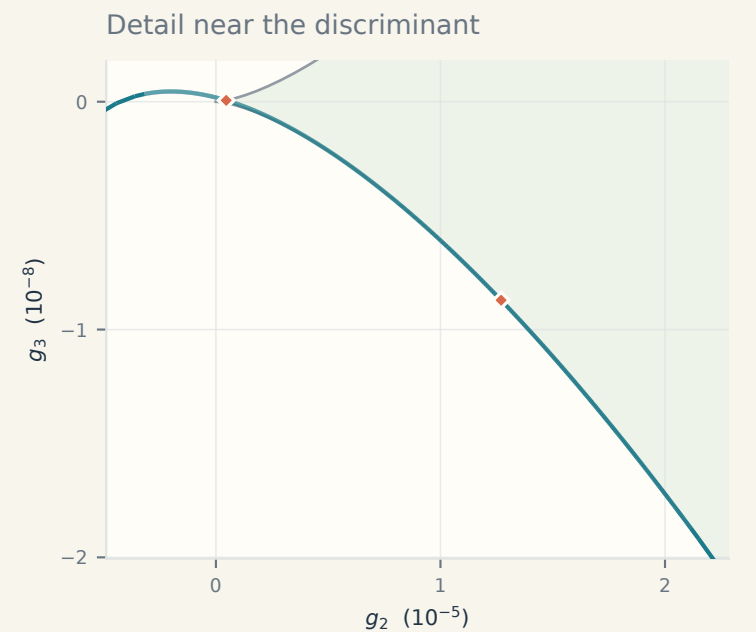
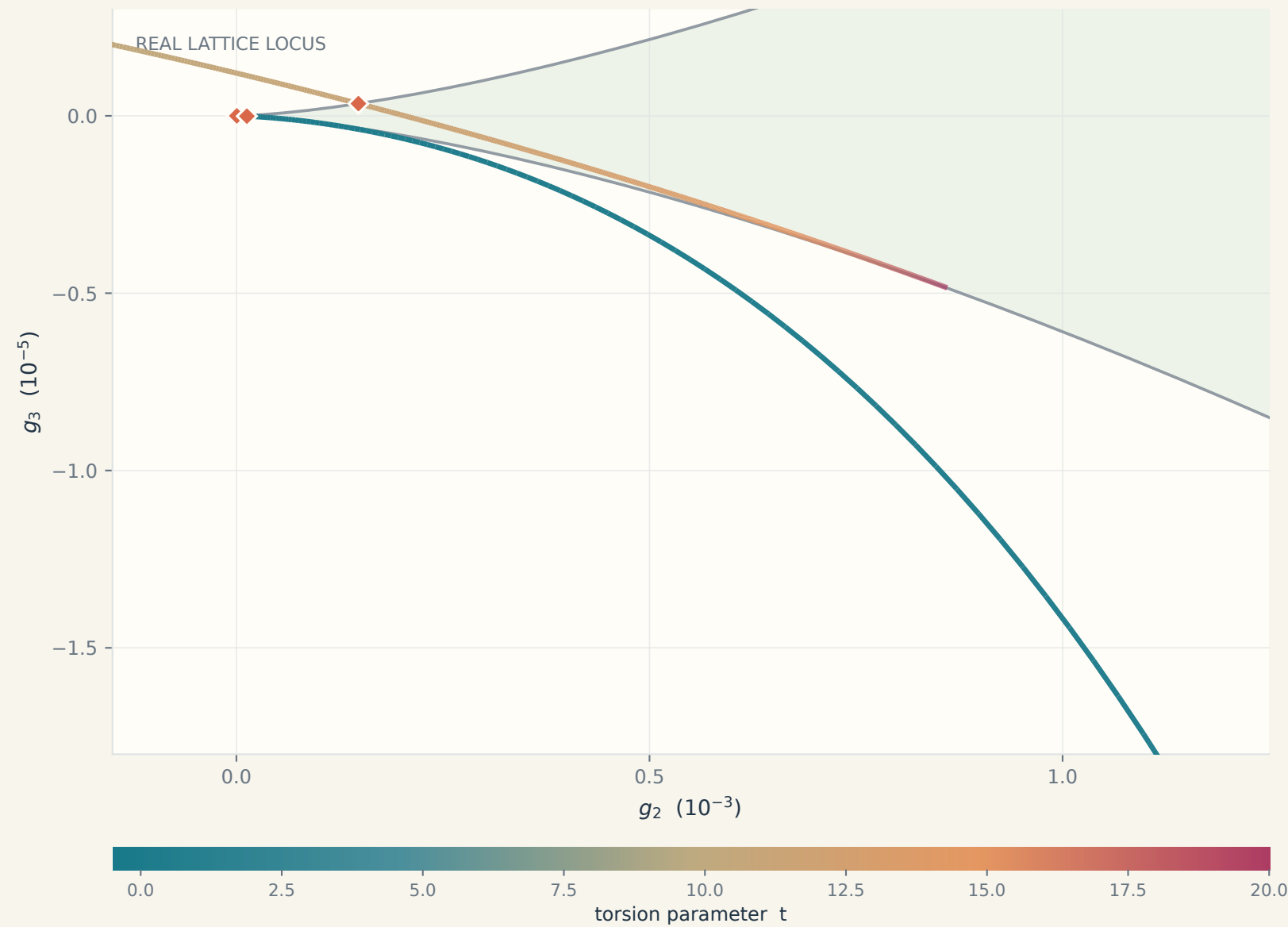
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

Every colored point gives a compatible equation; its remaining coefficients are read from the algebra.
Several points at the same a generally have different remaining coefficients, so they are different equations.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Sextic fifth order · a fivefold character

$(n, p) = (6, 5) \cdot q = 1$

$P(D)u + 120u^6 = 0, \quad a_4 = 1$



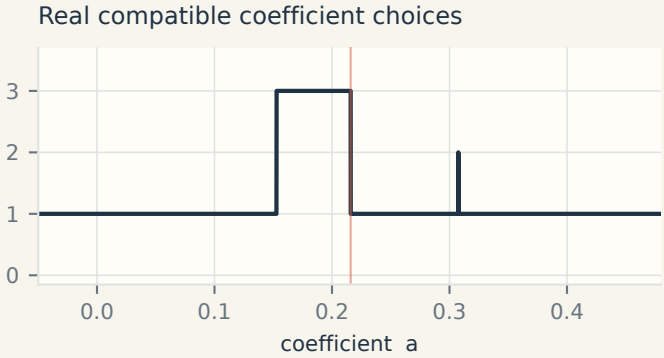
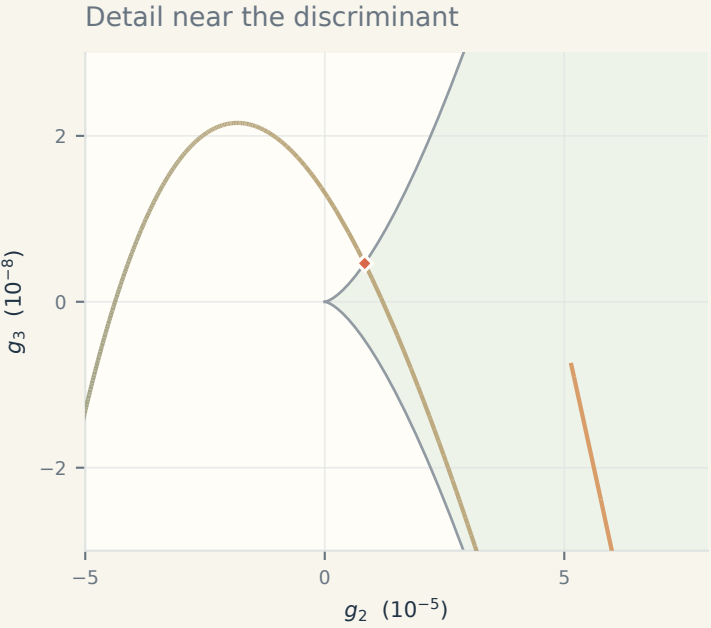
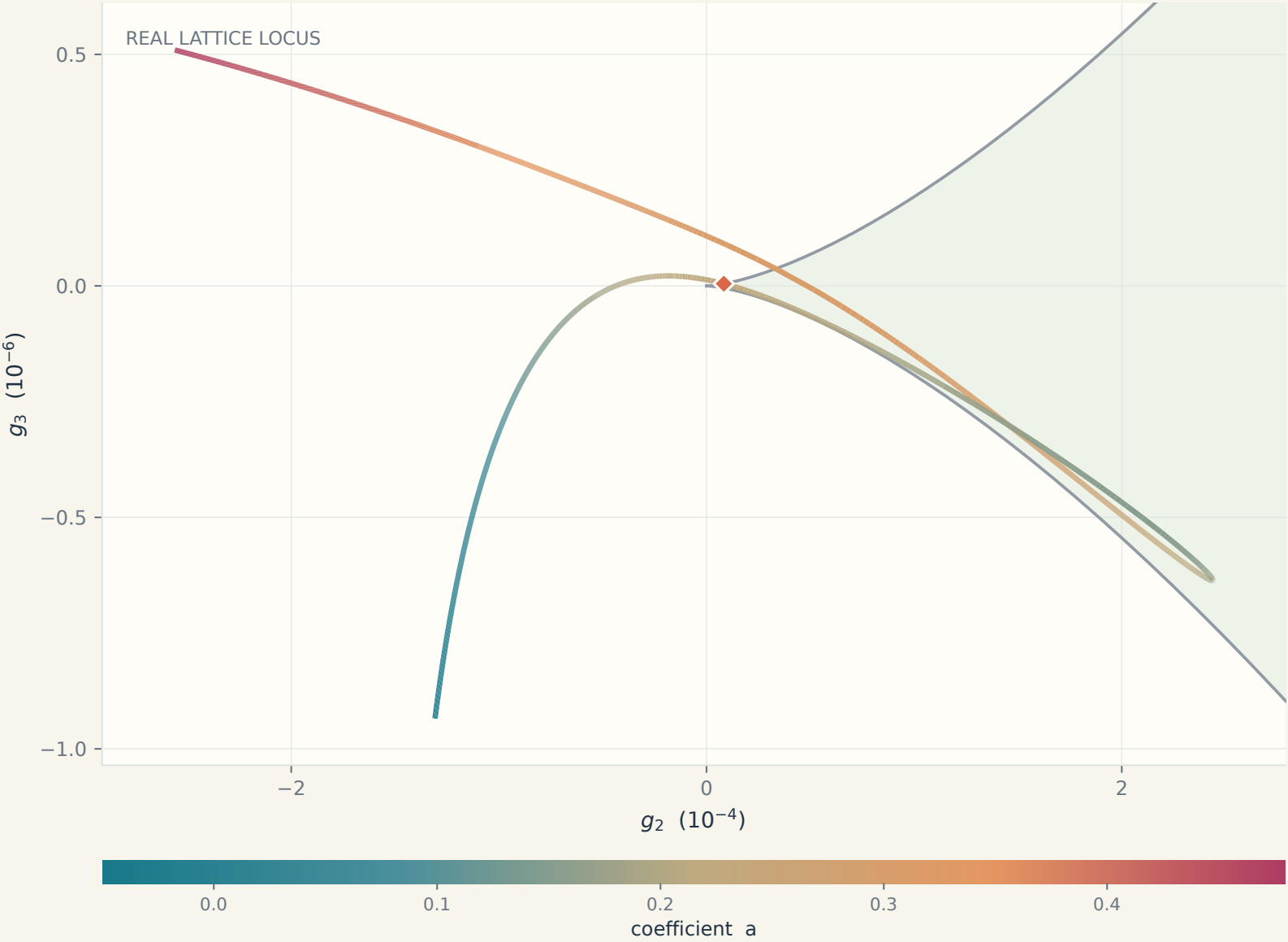
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

A character of order five gives five poles in an ordinary period cell; each displayed equation has one wave.
The chart fixes $a_4 = 1$. The value $t = 3$ has $a_4 = 0$ and is outside this normalization.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Quadratic sixth order · a three-branch fold

$(n, p) = (2, 6) \cdot q = 6$

$$u^{(6)} + u^{(4)} + au'' - 332640u^2 + C = 0$$



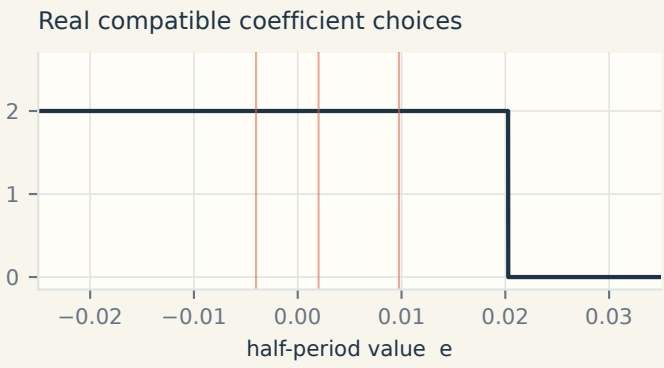
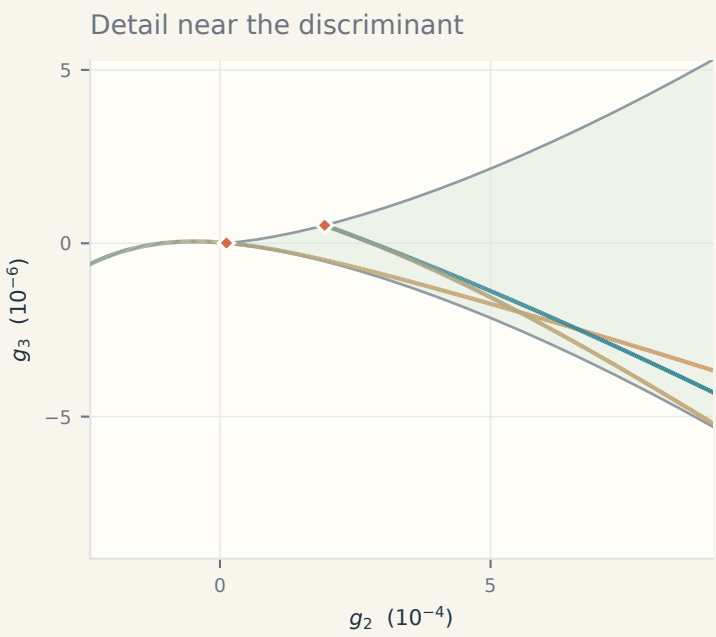
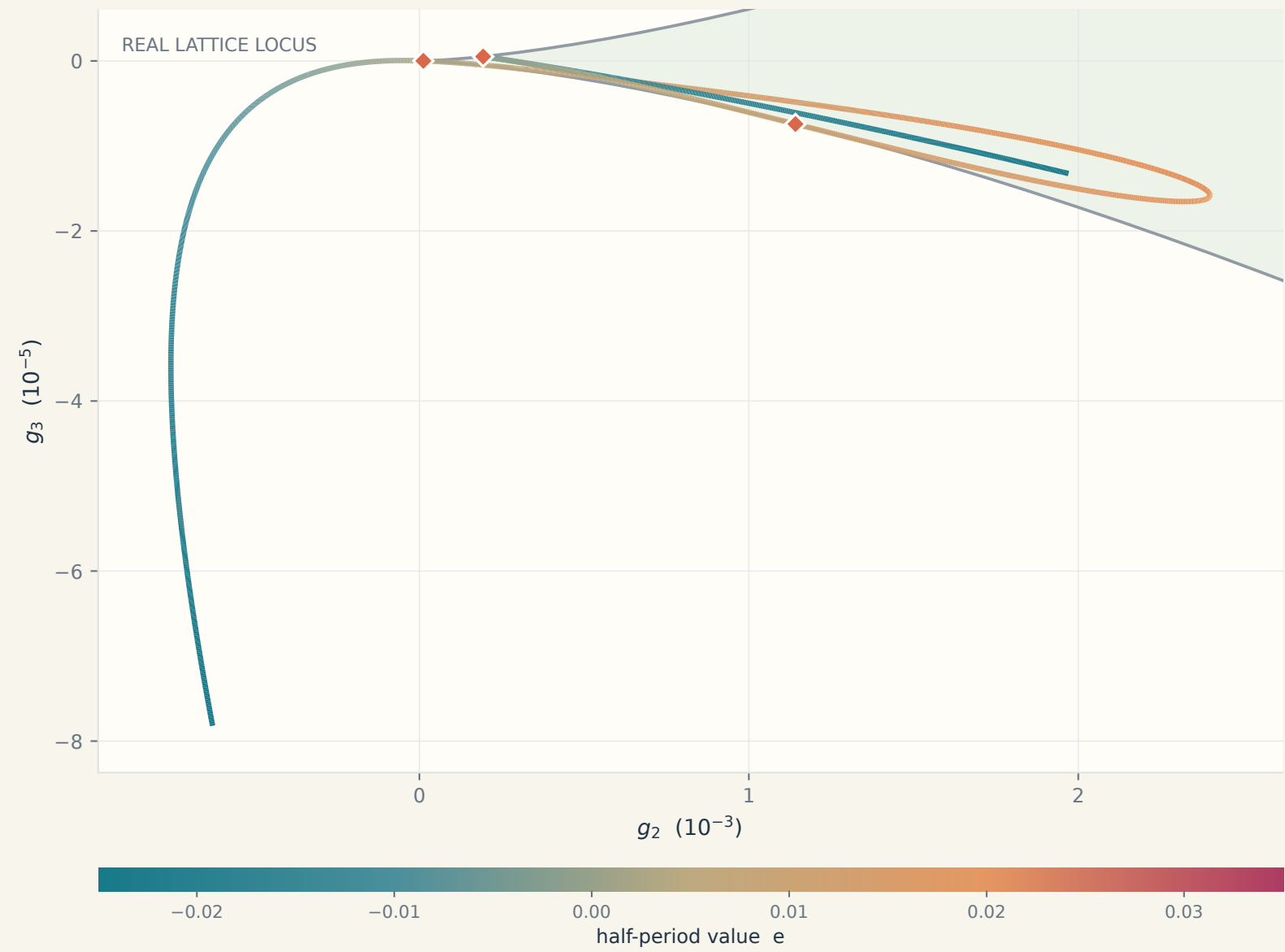
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

Every colored point gives a compatible equation; its remaining coefficients are read from the algebra.
Several points at the same a generally have different remaining coefficients, so they are different equations.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Cubic sixth order · a twisted bifurcation

$$(n, p) = (3, 6) \cdot q = 3$$

$$u^{(6)} + u^{(4)} + a_2 u'' + a_0 u - 20160 u^3 = 0$$



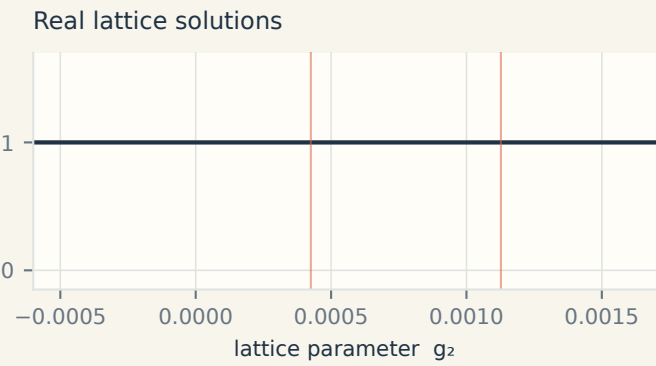
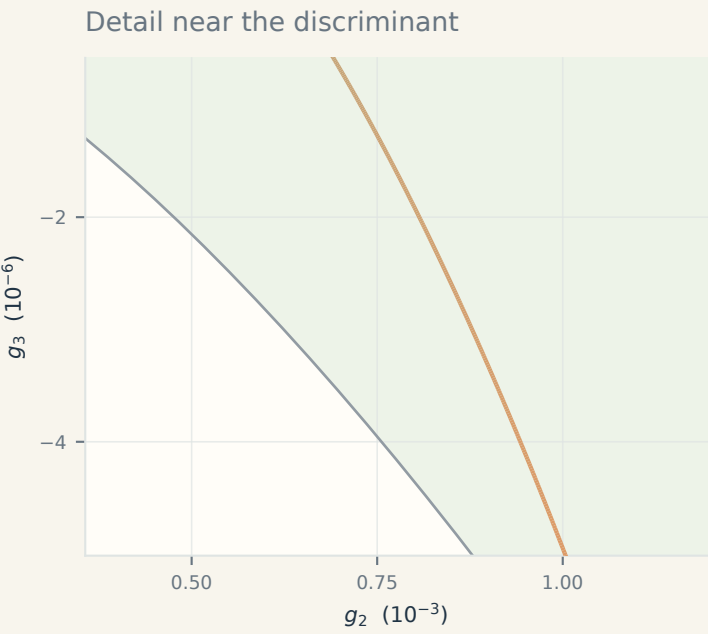
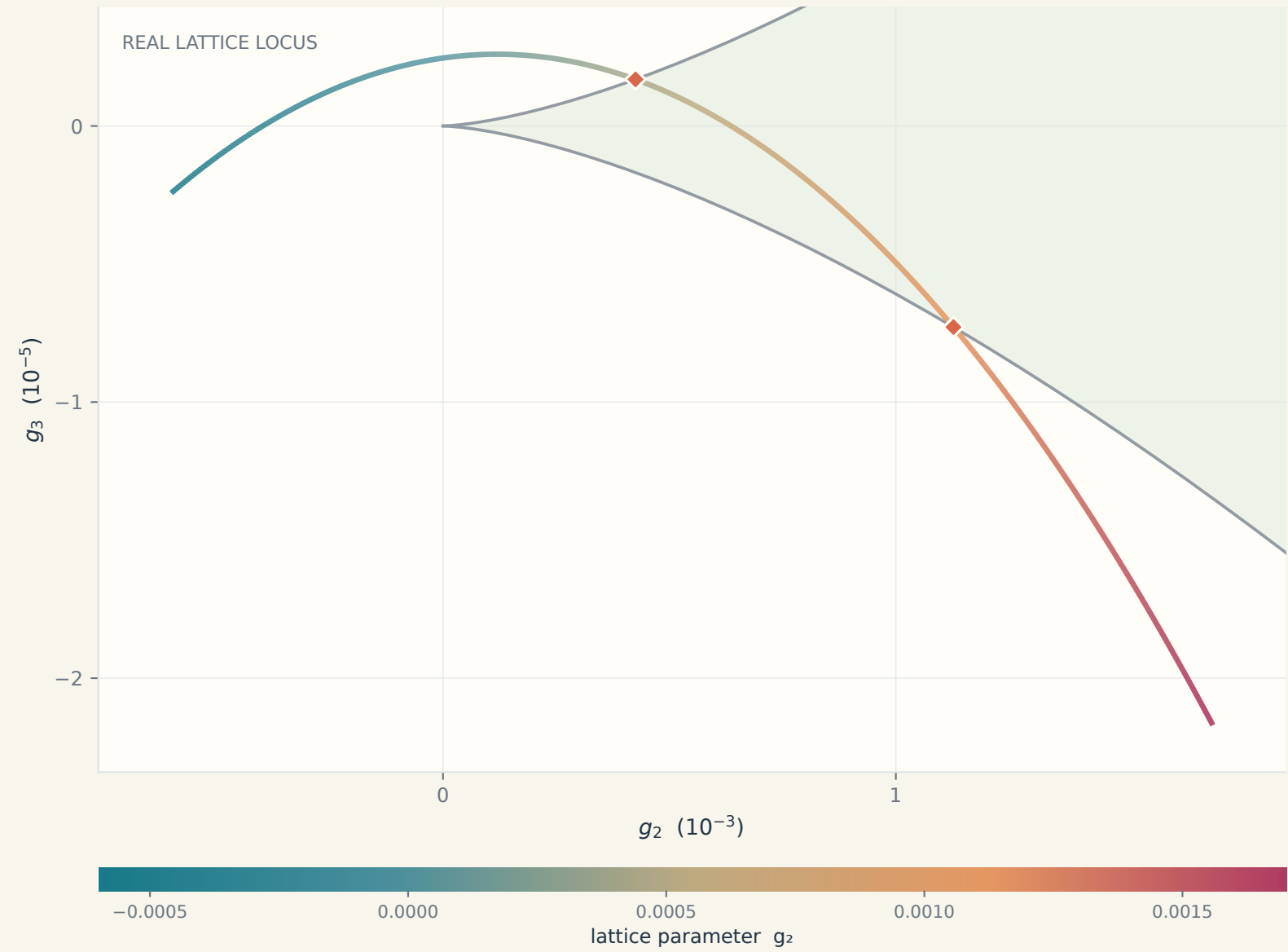
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

The two roots for g_2 give coefficient families $a_2(e)$, $a_0(e)$; the displayed branches can be different equations.
 $u = (\wp + (e + 1/83)/2)\sqrt{\wp - e}$. The ordinary branch requires $a_4 = 0$; see its separate plate.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Quartic sixth order · the parabolic branch

$(n, p) = (4, 6) \cdot q = 2$

$$u^{(6)} + u^{(4)} + a_2 u'' + a_0 u - 5040 u^4 = 0$$



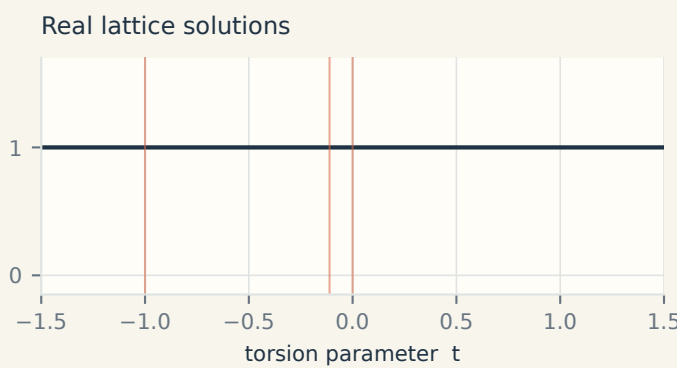
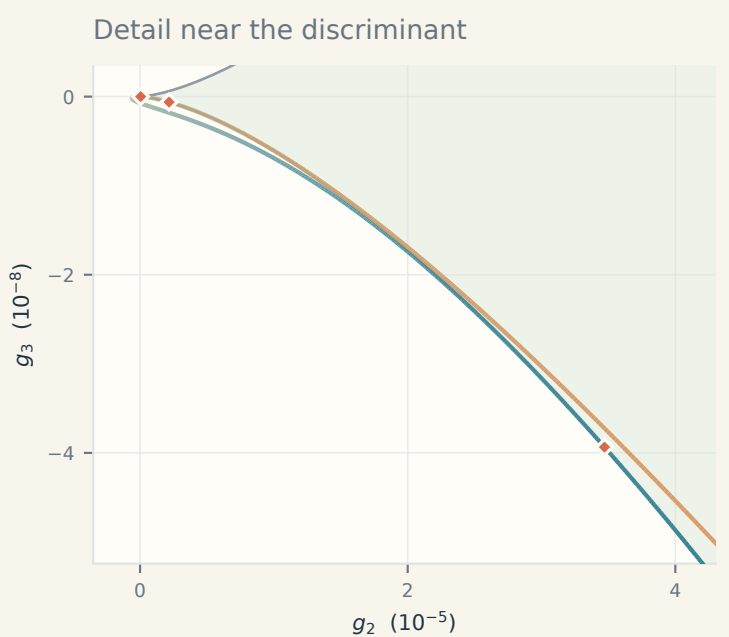
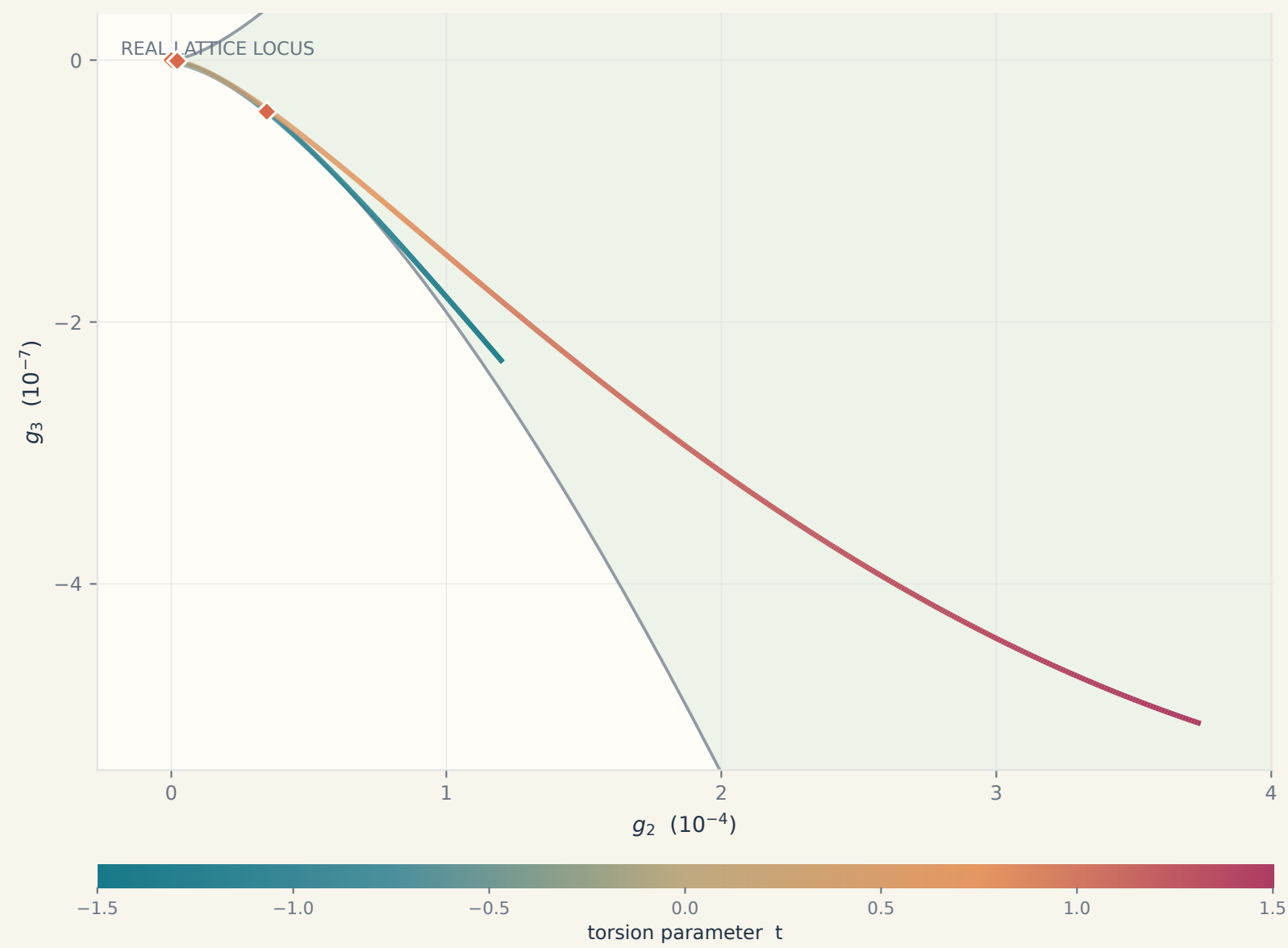
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

One wave per compatible equation; $u = \wp + 1/168$. The curve is an exact parabola in the lattice plane.
Here $a_2 = 168g_2 + 5/28$. The remaining coefficient a_0 is determined by the same substitution.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Septic sixth order · a sixfold character

$(n, p) = (7, 6) \cdot q = 1$

$P(D)u - 720u^7 = 0, \quad a_5 = 1$



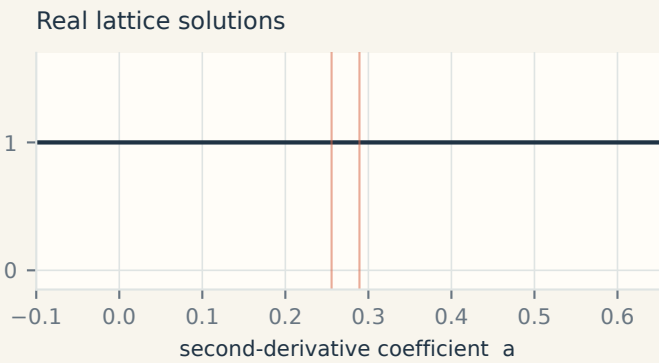
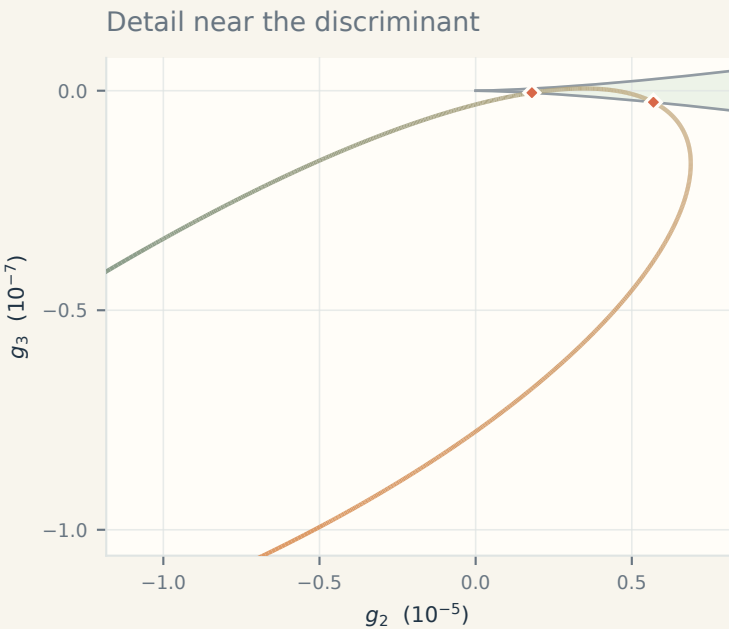
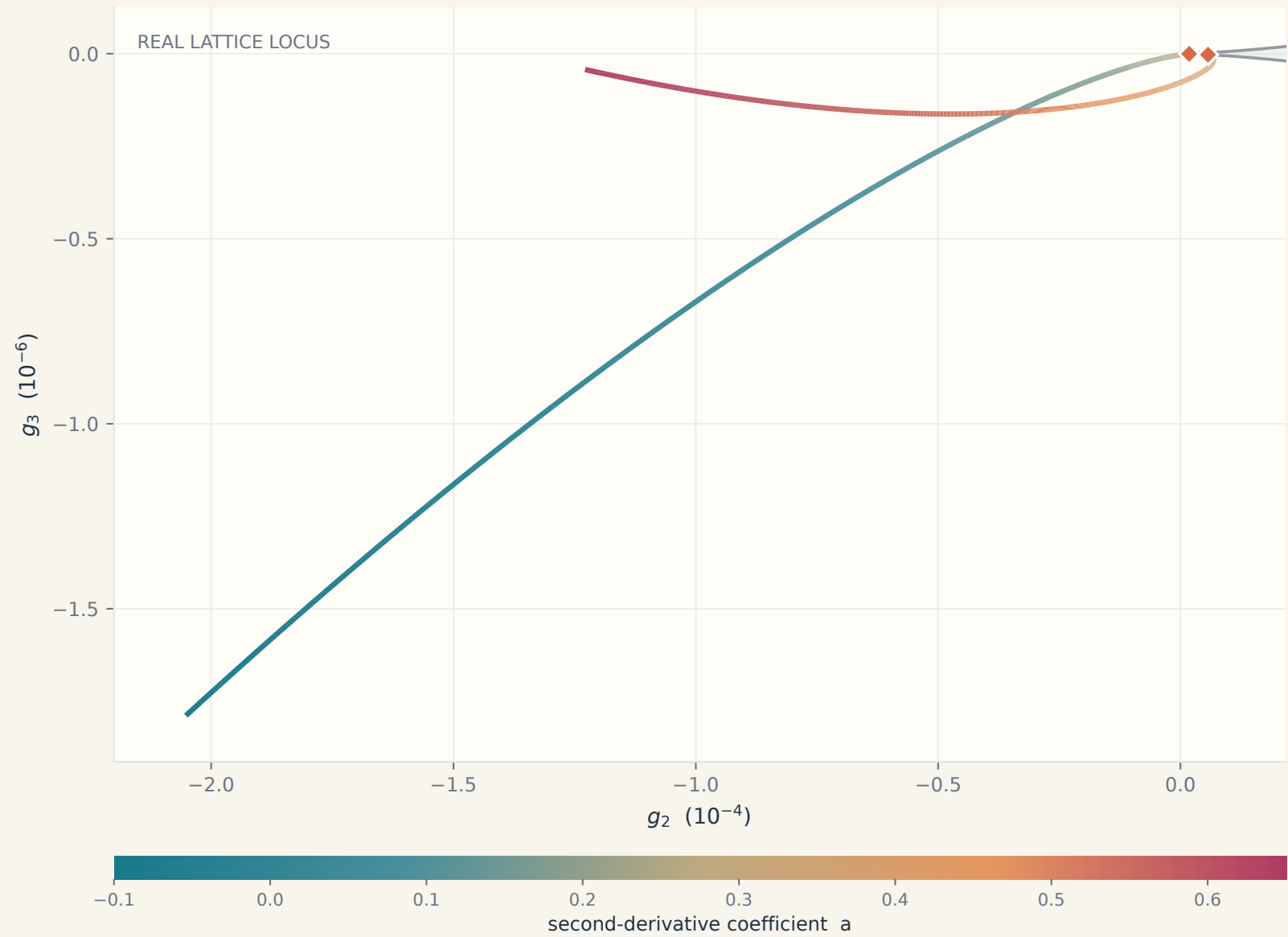
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

The normalized coefficient $a_5 = 1$ is fixed; the other coefficients follow an exact one-parameter family.
Characters of order six have six ordinary poles. The real cusp parameters are $t = -1, -1/9$, and 0 .
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Kawahara-KS · the dissipative curve

$(n, p) = (2, 4) \cdot q = 4$

$$u'''' + u''' + au'' + a_1(a)u' - 840u^2 + C(a) = 0$$



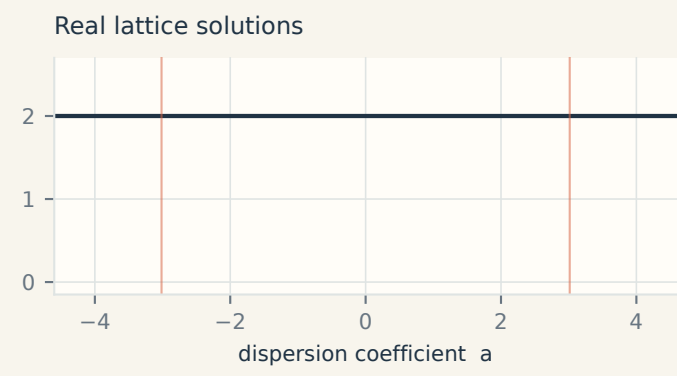
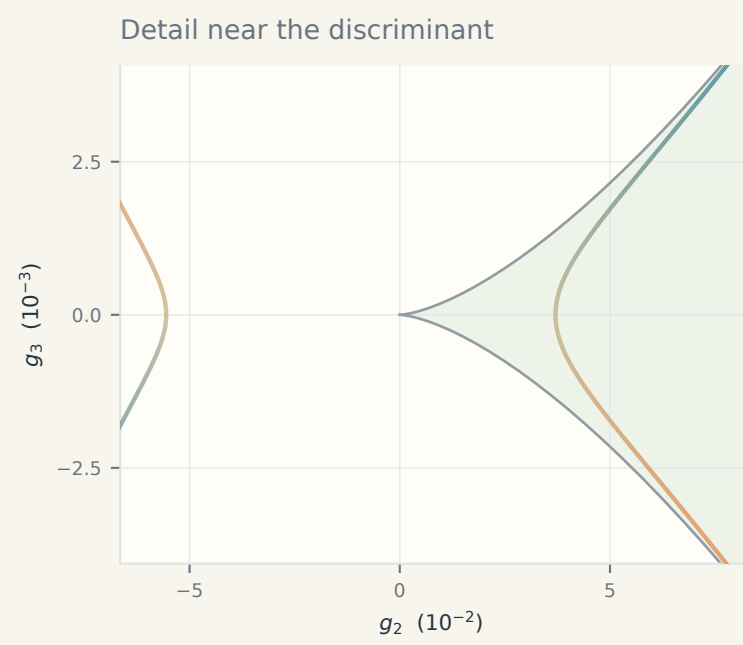
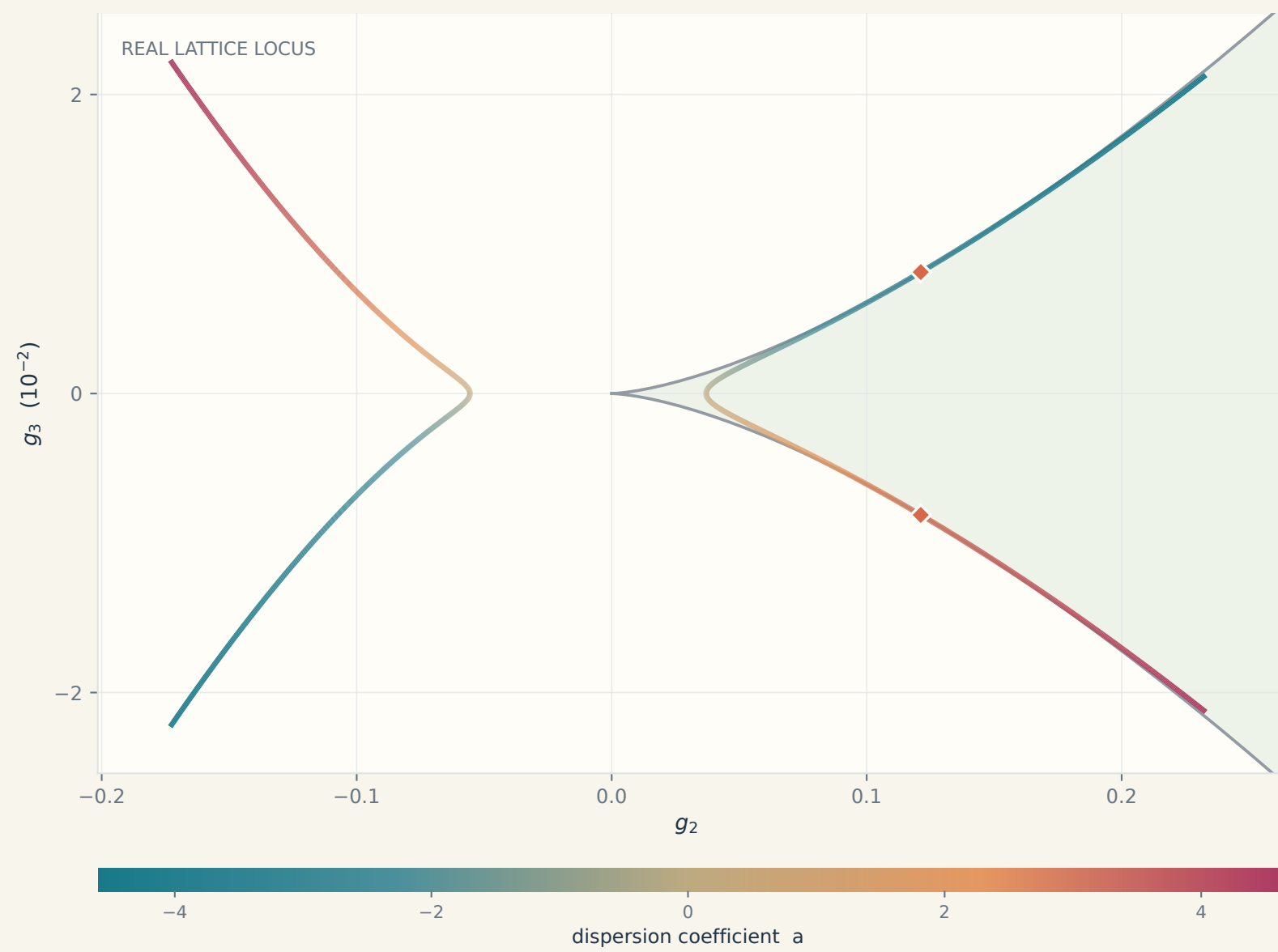
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

The odd derivative removes the two-wave freedom: one wave per compatible fixed equation.
The locus resolves several different passages through the discriminant.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Cubic fourth order · two tangencies

$(n, p) = (3, 4) \cdot q = 2$

$u'''' + au'' - u - 120u^3 = 0$



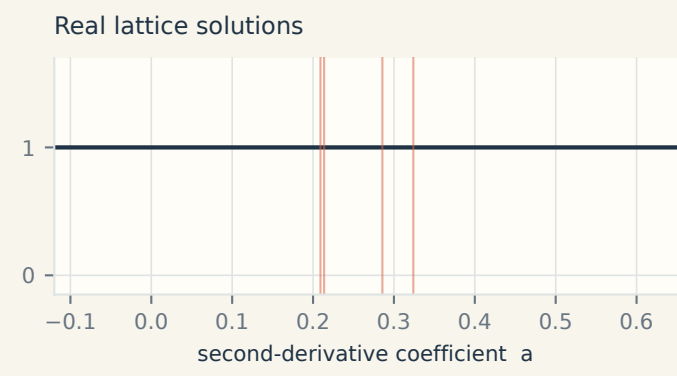
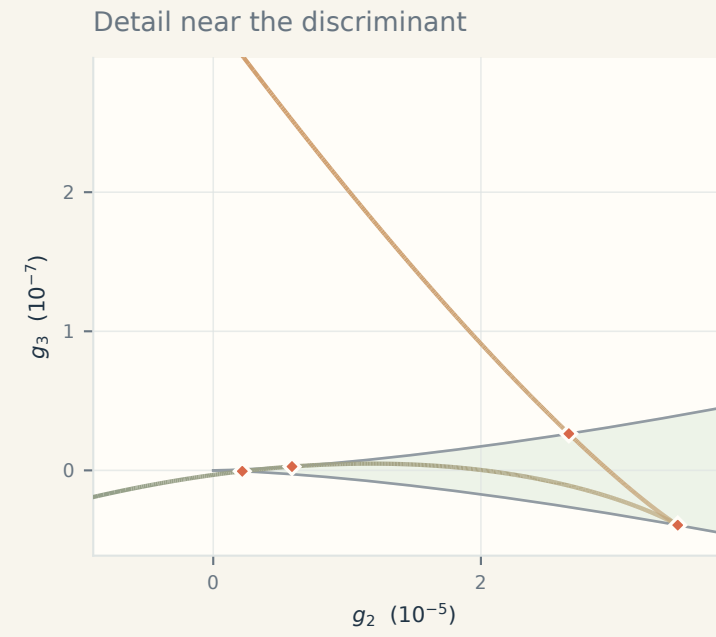
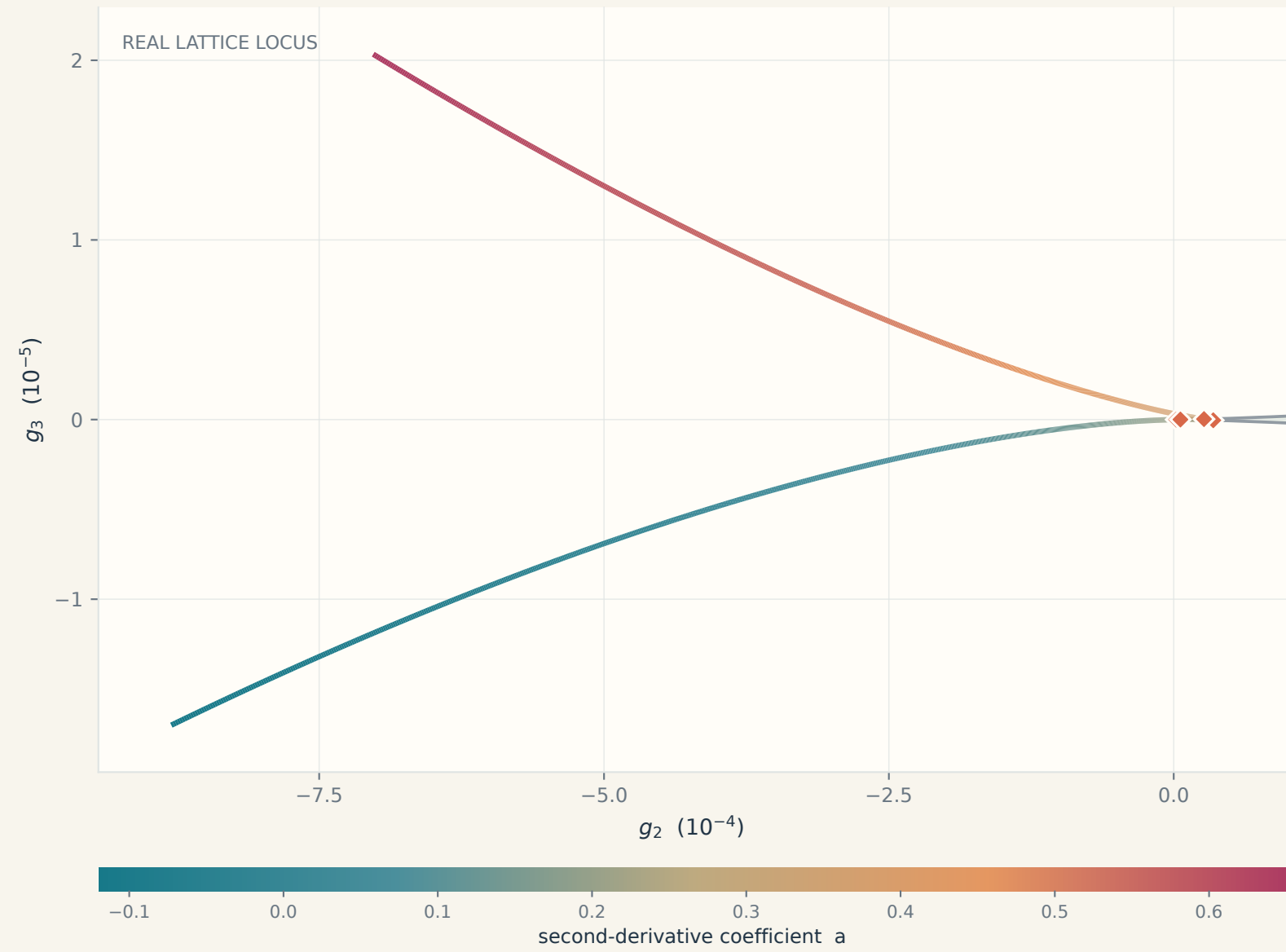
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

Both character branches are shown, up to translation and sign.
The twisted branch touches the cusp at $a = \pm 10/\sqrt{11}$; the ordinary branch stays at $\Delta < 0$.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Swift-Hohenberg · the dispersive curve

$(n, p) = (3, 4) \cdot q = 2$

$$u'''' + u''' + au'' + a_1(a)u' + a_0(a)u - 120u^3 = 0$$



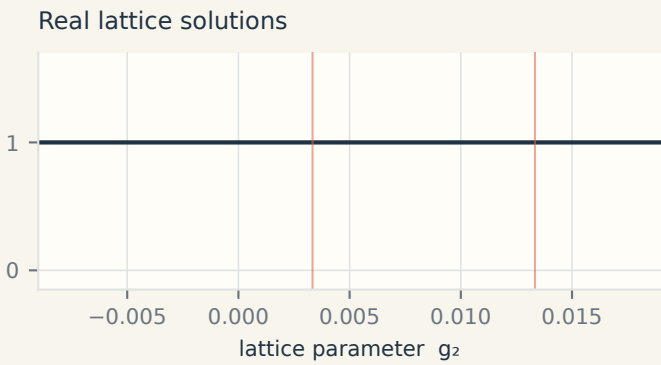
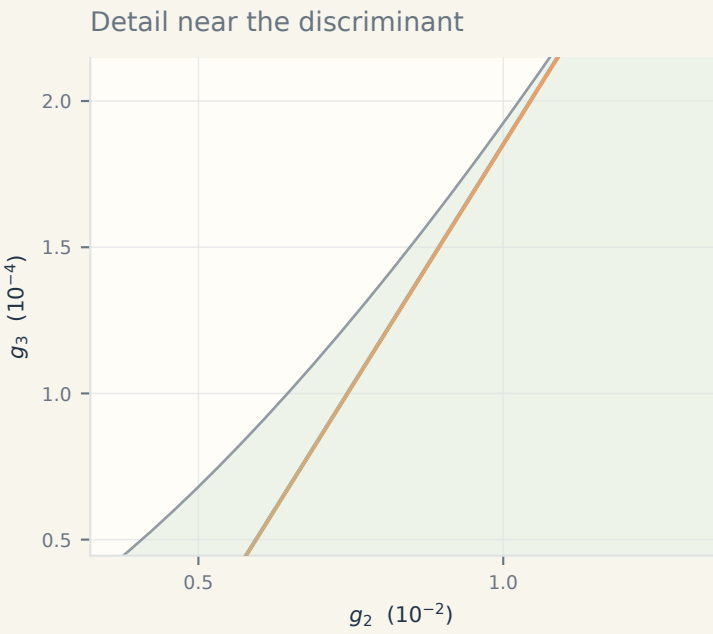
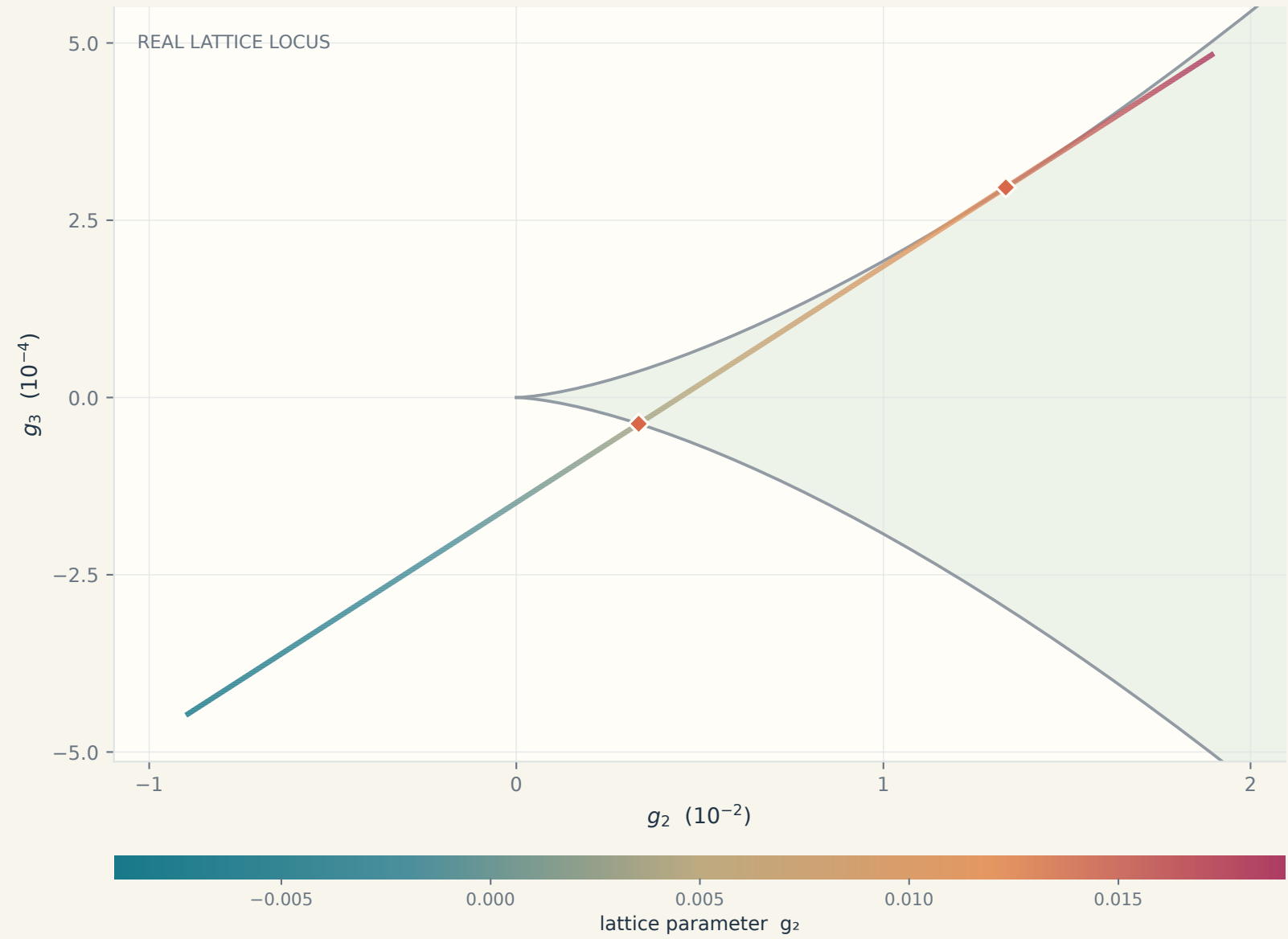
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

The residue condition fixes $a_1 = (196a - 39)/1372$, then a_0 and the lattice are fixed.
One wave per equation, with a character of order two.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Quintic · the even-operator line

$(n, p) = (5, 4) \cdot q = 1$

$$u^{(4)} + u^{(2)} + \cdots - 24u^5 = 0$$



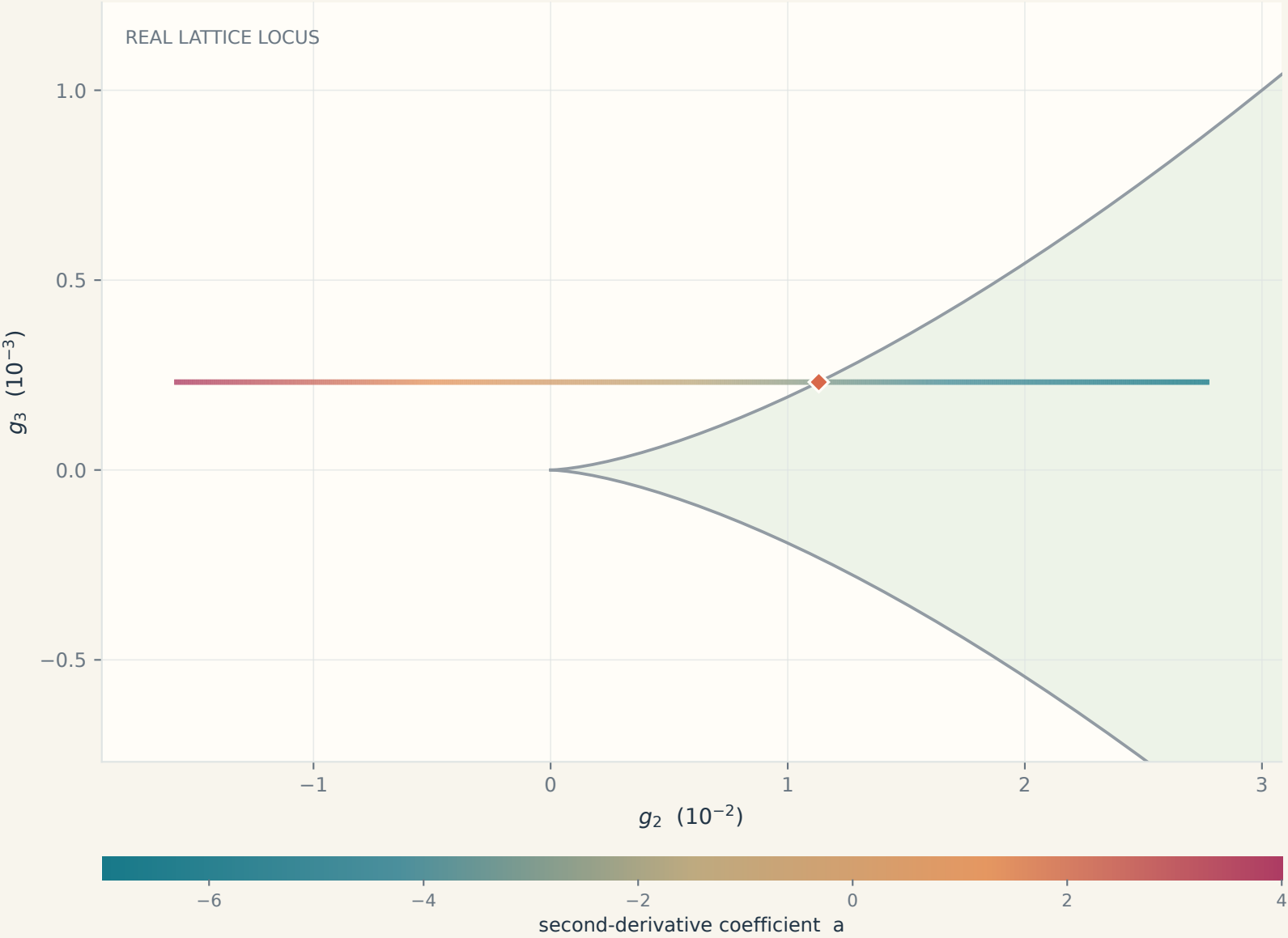
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

Even operators also carry characters of order two. This branch is separate from the full-order character family.
Every point determines the compatible lower coefficients; each such equation has one meromorphic wave.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

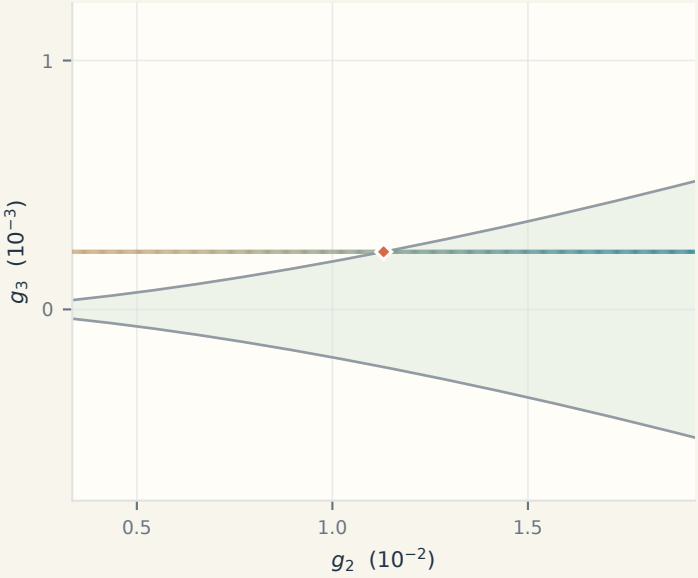
Cubic sixth order · the ordinary line

$(n, p) = (3, 6) \cdot q = 3$

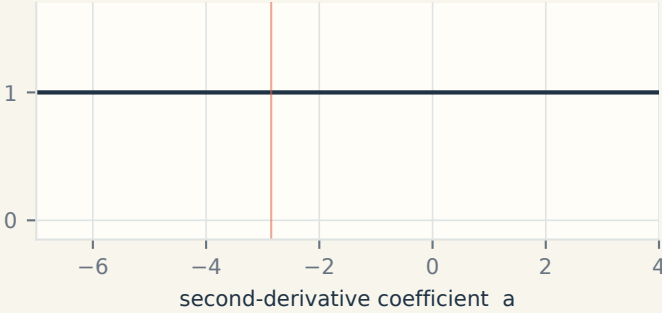
$u^{(6)} + au'' - u - 20160u^3 = 0$



Detail near the discriminant



Real lattice solutions



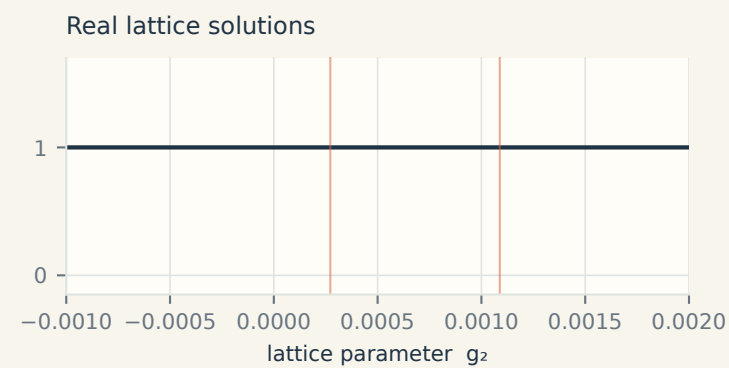
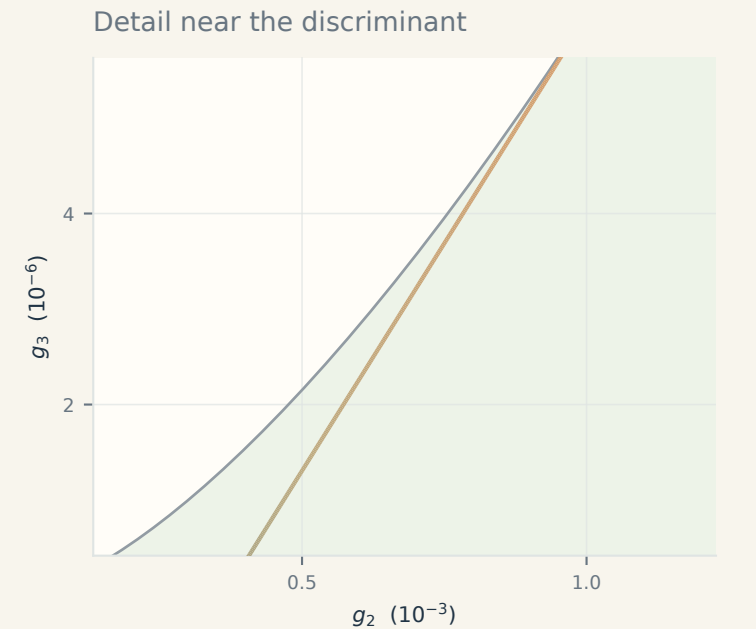
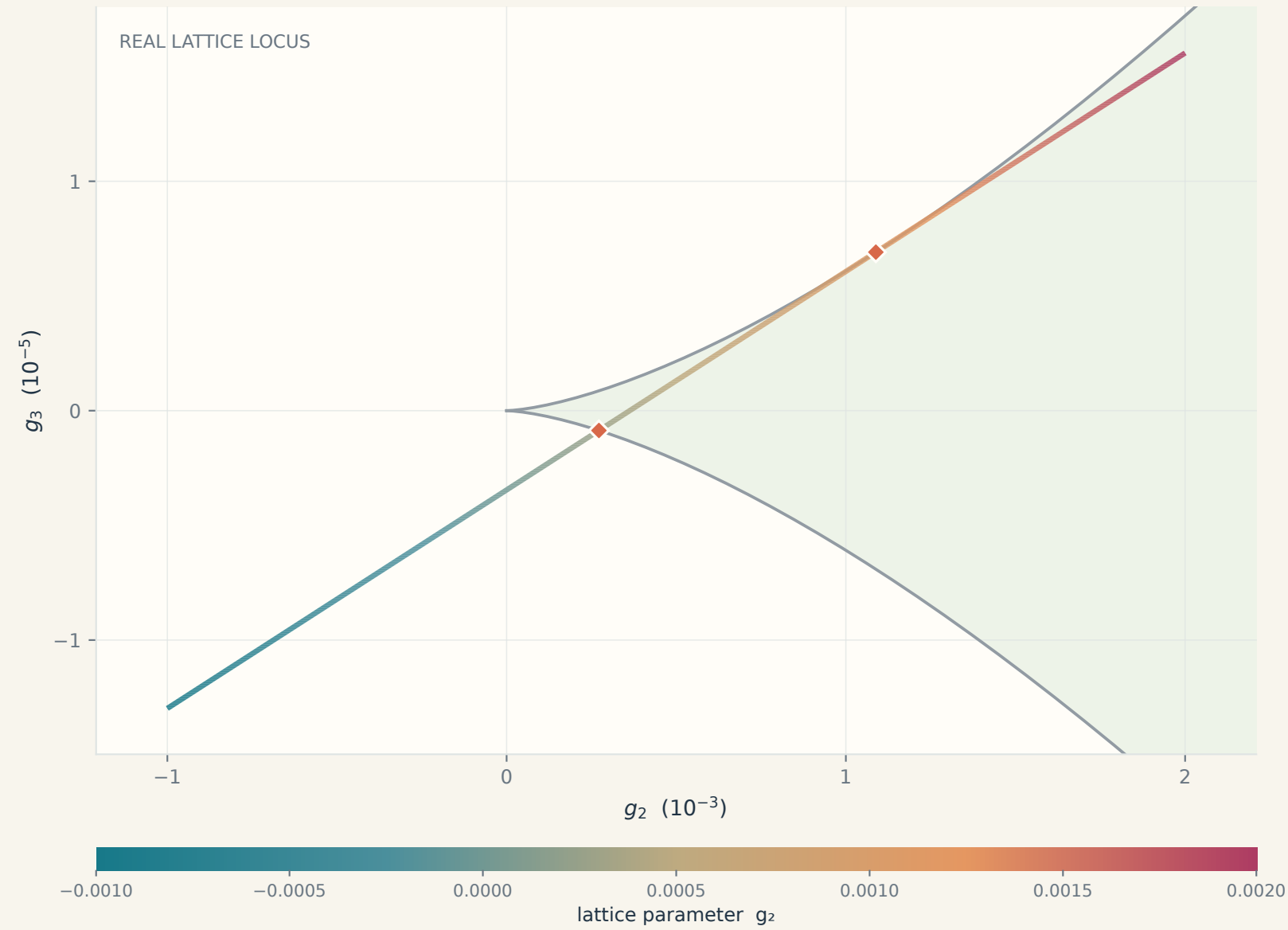
Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

With $a_4 = 0$ the elliptic wave is ordinary: $u = -p'/2$, one wave per fixed equation.
Its lattice follows a horizontal line; the pure $D^6 - 1$ equation is the point $g_2 = 0$.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

Septic · the even-operator line

$$u^{(6)} + u^{(4)} + \cdots - 720u^7 = 0$$

$$(n, p) = (7, 6) \cdot q = 1$$

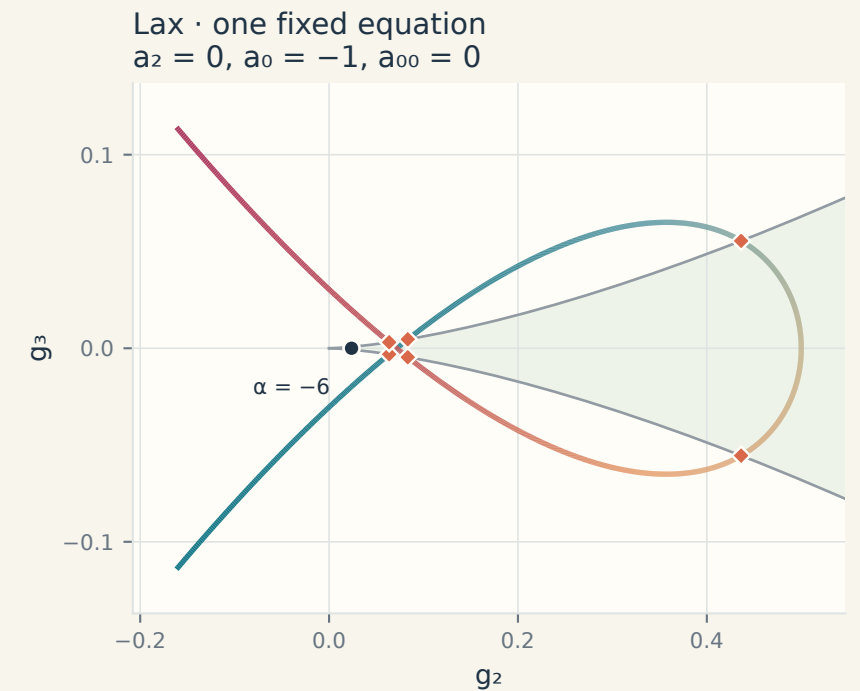
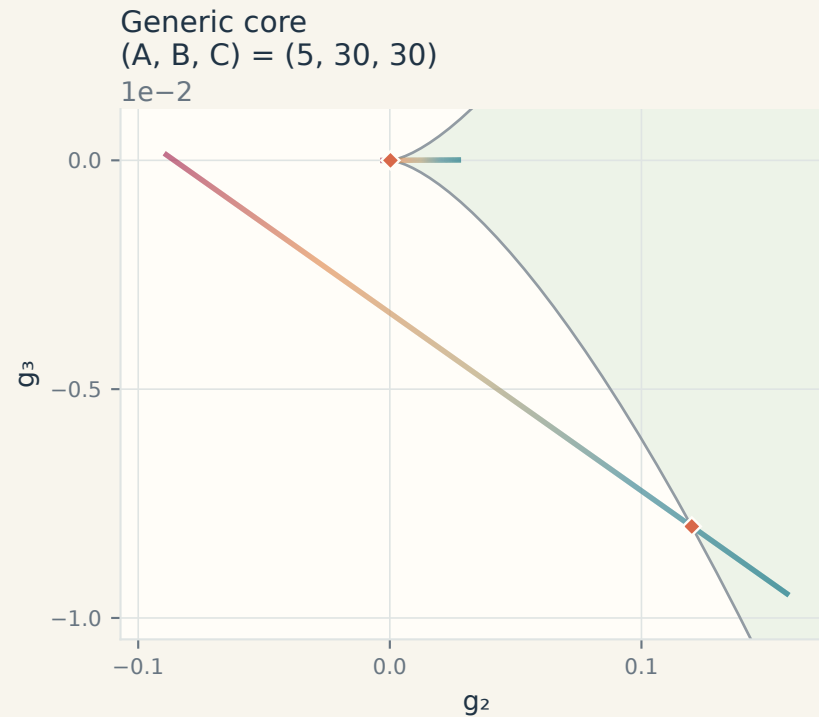
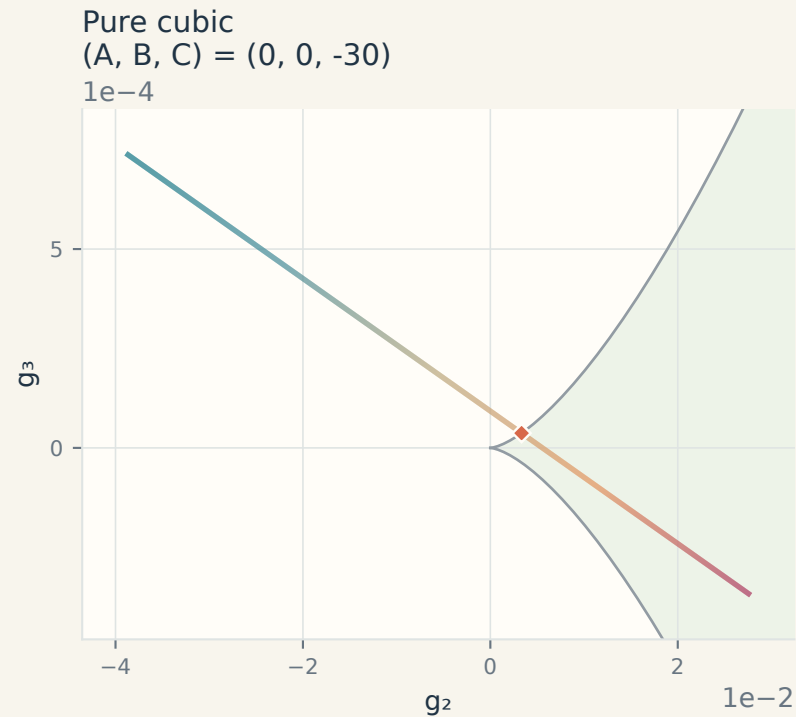
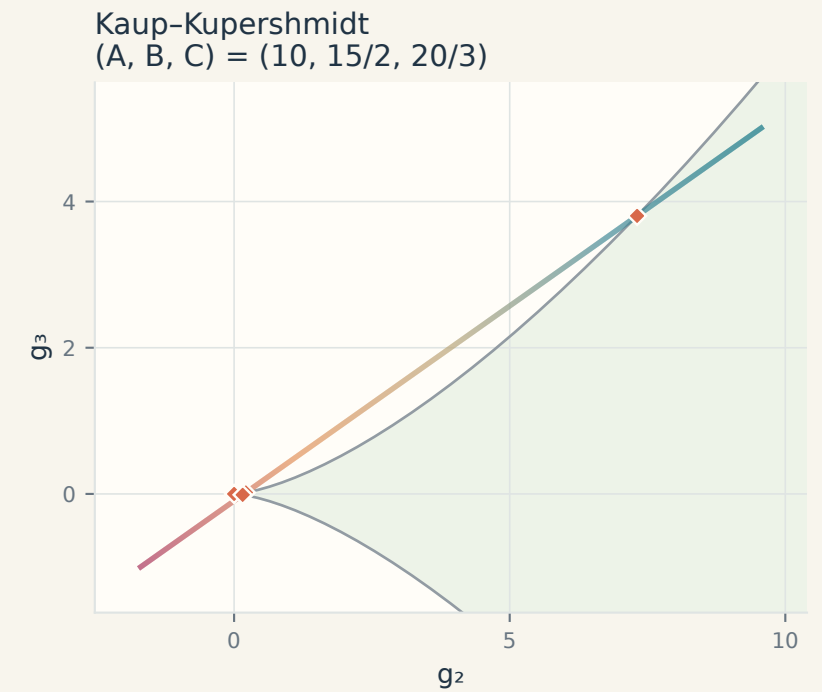
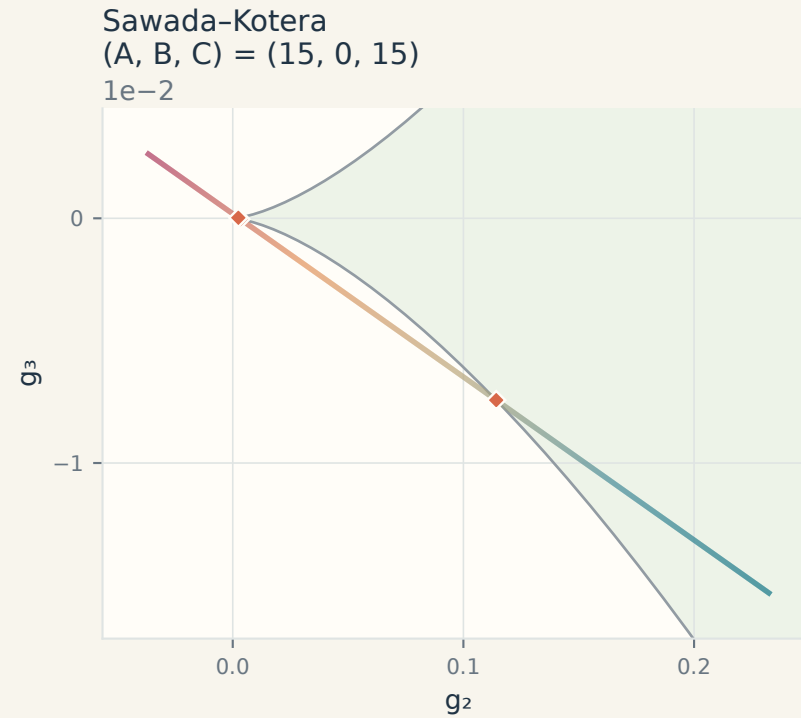
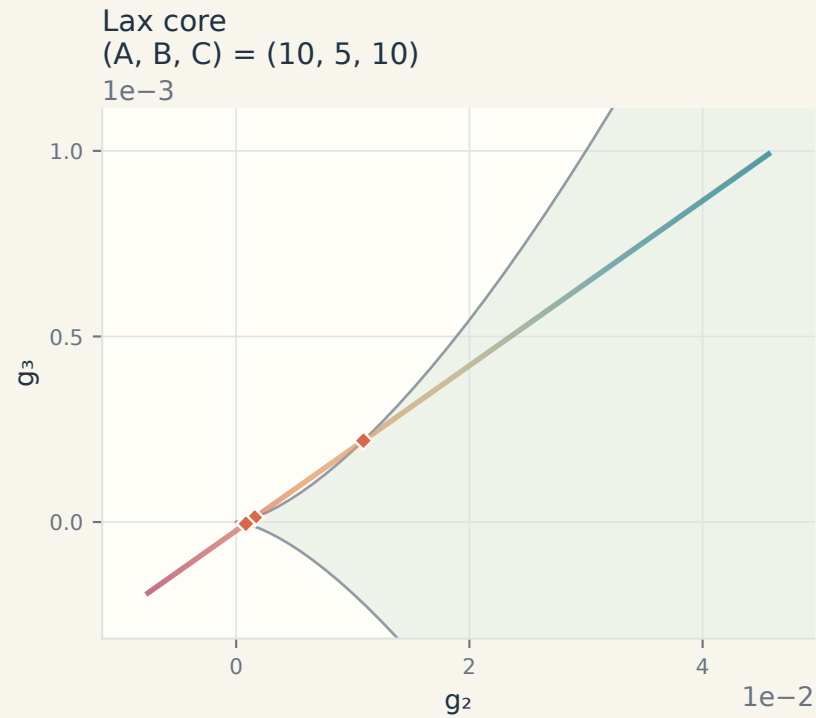


Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

Even operators also carry characters of order two. This branch is separate from the full-order character family.
Every point determines the compatible lower coefficients; each such equation has one meromorphic wave.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.

C4 · affine elliptic waves and a free lattice curve

$$v'''' + Avv'' + B(v')^2 + Cv^3 + a_2v'' + a_0v + a_{00} = 0, \quad v = \alpha\wp + \beta$$



Panels 1-5: $a_2 = 1, a_{00} = 0$; a_0 varies. Panel 6: one fixed Lax equation, with β free. Diamonds mark $\Delta = 0$.

The explorer lets you edit all six equation coefficients, select the sweep coordinate, focus a branch, and inspect its profile. Only the affine- $\wp$ sector is shown.